

Linear Algebra F*cked Up

Higher Algebra



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2026

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This is my personal notes on Higher Algebra.

0 Preliminaries

0.1 Pointed Simplicial Sets

0.1.1 Definition

Definition 0.1.1 (Pointed Simplicial Sets). The category of *pointed simplicial sets* is defined to be

$$\mathbf{sSet}_* := \mathbf{sSet}_{\Delta^0/} = (\Delta^0 \downarrow \mathbf{sSet})$$

where Δ^0 is the obviously the terminal object of $\mathbf{sSet}$.

Remark. Since for every n , there is a unique

$$\Delta^n \rightarrow \Delta^0$$

the base point of a pointed simplicial set $\Delta^0 \xrightarrow[x]{x}$ induces a constant n -simplex

$$(x_0)_n \in x_n, \quad \Delta^n \rightarrow \Delta^0 \rightarrow x$$

Geometrically, it is a n -simplex that is “collapsed” to the base point, with all its vertices identified with the base point.

0.1.2 Simplicial n -loops

Just like what we did in topology, we can define some “loops” in simplicial sets.

Definition 0.1.2 (Simplicial n -Loop). A *simplicial n -loop* of a pointed simplicial set (x, x_0) is a simplex $\sigma \in x_n$, where

$$d_i \sigma = (x_0)_{n-1}, \quad i = 0, \dots, n$$

which means that the border of σ degenerate to the base point.

Remark. Equivalently, the restriction of $\sigma : \Delta^n \rightarrow x$ to $\partial\Delta^n$ is the constant map at x_0 .

$$\begin{array}{ccc} \partial\Delta^n & \longrightarrow & \Delta^0 \\ \downarrow & & \downarrow x_0 \\ \Delta^n & \xrightarrow{\sigma} & x \end{array}$$

Because the border was collapsed to a point, σ factors uniquely through the quotient

$$\Delta^n / \partial\Delta^n \rightarrow x$$

where the notation $\Delta^n / \partial\Delta^n$ is somehow called the *n -sphere S^n* , defined by above. Hence we say

“simplicial n -loop $\iff$ pointed simplicial map $S^n \rightarrow x$ ”

0.2 Wedge Products and Smash Products

0.2.1 Wedge Products

Definition 0.2.3 (Wedge Product). Suppose $\Delta^0 \rightarrow x$ and $\Delta^0 \rightarrow y$ are two pointed simplicial sets, the *wedge product* of x and y is defined to be

$$x \vee y := x \sqcup_{\Delta^0} y$$

basically connecting the two base points together.

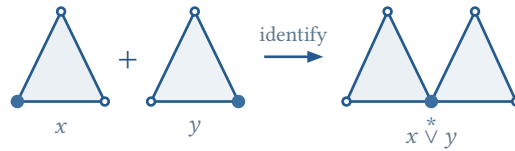


Figure 1: The wedge product identifies the marked base vertices of two pointed simplicial sets.

Remark. The wedge product is functorial in both variables, it is a bifunctor of 1-category

$$(- \vee -) : \mathbf{sSet}_* \times \mathbf{sSet}_* \rightarrow \mathbf{sSet}_*$$

also on the category of anima, this is also a ∞ -bifunctor

$$(- \vee -) : \mathbf{Ani}_* \times \mathbf{Ani}_* \rightarrow \mathbf{Ani}_*$$

0.2.2 Smash Products

Definition 0.2.4 (Smash Product). Suppose $\Delta^0 \rightarrow x$ and $\Delta^0 \rightarrow y$ are two pointed simplicial sets, the *smash product* of x and y is defined to be

$$x \wedge y := \frac{x \times y}{x \vee y}$$

basically collapsing the wedge product to a point.

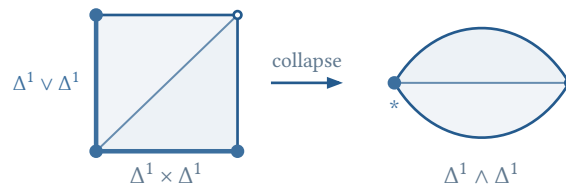


Figure 2: A minimal model of the smash product: the highlighted wedge in the triangulated product is collapsed to the marked vertex.

0.2.3 Some Theories on Wedge and Smash

Before discussing the formal properties, it is useful to keep the following analogy in mind: pointed spaces behave much like modules, with the base point playing the role of zero, the wedge product the role of direct sum, and the smash product the role of tensor product. This is a guide to intuition rather than an identification of the two settings.

R -modules	pointed spaces
0	$*$
$m + n$	$x \vee y$
$m \otimes_R n$	$x \wedge y$
$\text{Hom}_R(m, n)$	$\text{Map}_*(x, y)$

All the equivalences below are natural in the pointed simplicial sets involved. They can equally be read in the ∞ -category $\mathbf{Ani}_*$ of pointed anima.

Proposition 0.2.1 (Wedge Product Laws). The wedge product is symmetric and associative, and the one-point object $*$ is its unit:

$$\begin{aligned} x \vee y &\simeq y \vee x \\ (x \vee y) \vee z &\simeq x \vee (y \vee z) \\ x \vee * &\simeq x \end{aligned}$$

These equivalences express the fact that $x \vee y$ is the coproduct of x and y in the pointed category.

Proposition 0.2.2 (Smash Product Laws). Let $S^0 = \Delta^0 \sqcup \Delta^0$, with one summand chosen as the base point. The smash product is symmetric and associative, its unit is S^0 , and the one-point object is absorbing:

$$\begin{aligned} x \wedge y &\simeq y \wedge x \\ (x \wedge y) \wedge z &\simeq x \wedge (y \wedge z) \\ x \wedge S^0 &\simeq x \\ x \wedge * &\simeq * \end{aligned}$$

Proposition 0.2.3 (Distributivity and Tensor-Hom Adjunction). Smash product distributes over wedge product:

$$x \wedge (y \vee z) \simeq (x \wedge y) \vee (x \wedge z).$$

Moreover, pointed mapping spaces satisfy the natural adjunction

$$\text{Map}_*(x \wedge y, z) \simeq \text{Map}_*(x, \text{Map}_*(y, z)).$$

Thus $x \wedge (-)$ preserves colimits, which explains the distributive law.

Proof. Write x_0 , y_0 , and z_0 for the chosen base points. By definition, the quotient map

$$q : x \times y \rightarrow x \wedge y$$

collapses $x \vee y = (x \times \{y_0\}) \cup (\{x_0\} \times y)$ to the base point. Hence a pointed map $g : x \wedge y \rightarrow z$ is the same thing as a map $f : x \times y \rightarrow z$ satisfying

$$f(x, y_0) = z_0 = f(x_0, y)$$

for every x and y .

Currying f gives a map

$$x \rightarrow \text{Map}(y, z), \quad x \mapsto (y \mapsto f(x, y))$$

The condition $f(x, y_0) = z_0$ says that every map $y \rightarrow z$ in its image is pointed, so this map factors through $\text{Map}_*(y, z)$. The condition $f(x_0, y) = z_0$ says that x_0 is sent to the constant base-point map, so the curried map $x \rightarrow \text{Map}_*(y, z)$ is itself pointed. This construction is reversible and natural. Applied degreewise, it gives the equivalence of pointed mapping spaces

$$\text{Map}_*(x \wedge y, z) \simeq \text{Map}_*(x, \text{Map}_*(y, z))$$

By symmetry, $x \wedge (-)$ is therefore left adjoint to $\text{Map}_*(x, -)$ and preserves all colimits. Since $y \vee z$ is the coproduct of y and z in the pointed category, preservation of this coproduct gives

$$x \wedge (y \vee z) \simeq (x \wedge y) \vee (x \wedge z)$$

as claimed. □

In other words, $(\mathbf{sSet}_*, \wedge, S^0)$ is a closed symmetric monoidal category, with internal Hom object $\text{Map}_*(-, -)$. Likewise, $(\mathbf{Ani}_*, \wedge, S^0)$ is a closed symmetric monoidal ∞ -category.

Remark (Spheres and Suspension). For pointed spheres one obtains

$$S^m \wedge S^n \simeq S^{m+n}$$

the proof is obvious and direct.

0.3 Suspension and Looping

0.3.1 Suspension

Intuitively, the suspension of a pointed simplicial set is obtained by taking the product with a simplicial interval and collapsing the two ends to points. Formally, we have

Definition 0.3.5 (Suspension). The *suspension* of a pointed simplicial set x is defined to be

$$\Sigma x := S^1 \wedge x$$

0.3.2 Looping

According to the “ n -loop = pointed map from n -sphere” principle, we have

Definition 0.3.6 (Loop). The *loop* of a pointed simplicial set x is defined to be

$$\Omega x := \text{Map}_*(S^1, x)$$

0.3.3 Some theories on Suspension and Looping

Theorem 0.3.1 (Suspension-Loop Adjunction). There is a natural equivalence of pointed mapping spaces

$$\mathrm{Map}_*(\Sigma x, y) \simeq \mathrm{Map}_*(x, \Omega y)$$

for every pointed simplicial sets x and y . In other words, there is an adjunction

$$\Sigma : \mathbf{sSet}_* \rightleftarrows \mathbf{sSet}_* : \Omega$$

Proof. Immediately by definition and by Proposition 0.2.3. □

0.3.4 Module Analogy

Let R be a commutative ring. For a fixed R -module ℓ , the ordinary tensor–Hom adjunction is

$$\mathrm{Hom}_R(\ell \otimes_R m, n) \simeq \mathrm{Hom}_R(m, \mathrm{Hom}_R(\ell, n)).$$

Thus the pointed-space dictionary extends from individual operations to adjoint functors:

$$\begin{aligned} k \wedge (-) &\leftrightarrow \ell \otimes_R (-) \\ \mathrm{Map}_*(k, -) &\leftrightarrow \mathrm{Hom}_R(\ell, -). \end{aligned}$$

Taking $k = S^1$ on the space side gives $\Sigma \dashv \Omega$. The closer algebraic analogue appears in the derived category $D(R)$, where the shifted unit $R[1]$ satisfies

$$\begin{aligned} R[1] \otimes_R^L m &\simeq m[1] \\ \mathrm{RHom}_R(R[1], n) &\simeq n[-1]. \end{aligned}$$

Hence the shift $[1]$ is left adjoint to $[-1]$, and the two are inverse equivalences. This last feature belongs to the stable setting: after pointed spaces are stabilized to spectra, Σ and Ω likewise become inverse equivalences; in $\mathbf{Ani}_*$ they are adjoint but not inverse in general. We can even claim ambitiously

“ Stable homotopy theory = Linear algebra over the sphere spectrum $\mathbb{S}$. ”

We will explore this analogy in further chapters.

0.4 Homotopy Groups

The simplicial n -loops introduced earlier are individual representatives. To obtain an invariant of the homotopy type of x , we identify representatives that are homotopic through pointed maps.

Definition 0.4.7 (Homotopy Groups). Let (x, x_0) be a pointed anima, represented for example by a pointed Kan complex. For every $n \geq 0$, define

$$\pi_n(x, x_0) := \pi_0 \mathrm{Map}_*(S^n, x) = [S^n, x]_*,$$

where $[S^n, x]_*$ denotes the set of pointed homotopy classes of maps. In terms of iterated loop spaces, the same definition is

$$\pi_n(x, x_0) = \pi_0(\Omega^n x),$$

where $\Omega^0 x = x$ and the base point of $\Omega^n x$ is the constant loop at x_0 .

Remark (Why a Kan Replacement Is Needed). If x is an arbitrary pointed simplicial set, the expression above means the *derived* pointed mapping space. Concretely, choose a pointed Kan fibrant replacement $x \rightarrow x^{\text{fib}}$ and set

$$\pi_n(x, x_0) := \pi_0 \text{Map}_*(S^n, x^{\text{fib}}).$$

This makes π_n invariant under weak equivalence. When x is already Kan, no replacement is necessary.

Remark (Simplicial Description). For a pointed Kan complex (x, x_0) , a map $S^n = \Delta^n / \partial \Delta^n \rightarrow x$ is exactly an n -simplex $\sigma \in x_n$ satisfying

$$d_i \sigma = (x_0)_{n-1}, \quad 0 \leq i \leq n.$$

Therefore $\pi_n(x, x_0)$ is the set of such simplicial n -loops modulo pointed simplicial homotopy relative to $\partial \Delta^n$. The quotient by homotopy is essential: the raw set of n -loops is not itself a homotopy invariant.

Proposition 0.4.4 (Group Structure). For $n \geq 1$, $\pi_n(x, x_0)$ is a group; for $n \geq 2$, it is abelian. The product of two classes $[f]$ and $[g]$ is represented by the composite

$$S^n \xrightarrow{\text{pinch}} S^n \vee S^n \xrightarrow{f \vee g} x.$$

Equivalently, this is concatenation of components in the n -fold loop space $\Omega^n x$. The identity is the constant map at x_0 .

Proof sketch. The loop space $\Omega^n x$ has concatenation, so its connected components form a group when $n \geq 1$. For $n \geq 2$, the two loop directions give two compatible multiplications; the Eckmann–Hilton argument makes the product commutative. $\square$

0.4.1 The Whitehead Theorem

Theorem 0.4.2 (Whitehead Theorem). Suppose $f : x \rightarrow y$ is a map of anima. Suppose

$$\pi_0(x) \xrightarrow{\sim} \pi_0(y), \quad \pi_n(x, x_0) \xrightarrow{\sim} \pi_n(y, f(x_0))$$

holds for every base point x_0 . Then f is a categorical equivalence.

1 Stable ∞ -Categories

1.1 The Motivation of the Subject

Ordinary category theory remembers a *set* of maps between two objects. This is often enough in classical algebra, but it is too small for homotopy theory and derived mathematics. There, two maps may be connected by a homotopy, two homotopies may themselves be homotopic, and these higher relations affect later constructions. Higher Algebra keeps this information instead of collapsing it.

Localization. When quasi-isomorphisms or weak equivalences are inverted, an ordinary localization remembers only homotopy classes of maps. An ∞ -categorical localization retains the whole mapping space. Its truncation is

$$\mathrm{Hom}_{hC}(x, y) = \pi_0 \mathrm{Map}_C(x, y)$$

Thus the ordinary derived category is useful, but it is only a shadow of the enhanced object.

Coherence. “Associative up to homotopy” is not enough by itself. The chosen homotopies must agree with one another, their agreements must also agree, and so on. A_∞ - and E_n -algebras, encoded by ∞ -operads, package this infinite system of compatibility conditions.

Stability. Stable ∞ -categories provide functorial fibers, cofibers, shifts, and mapping spectra. They unify derived categories and spectra while retaining the information that a triangulated homotopy category forgets.

Remark (Worldview). Higher Algebra replaces “unique by equality” with “a contractible space of coherent choices.” Its guiding principle is simple: keep enough structure so that the next construction is invariant, functorial, and composable.

In short, ∞ -categories retain higher morphisms, stable ∞ -categories handle derived and exact phenomena, and ∞ -operads describe algebraic operations up to every required coherence. These tools are used because the mathematics is already higher, not because abstraction is an end in itself.

1.2 First Definitions: Stability

1.2.1 Pointed ∞ -Categories

Definition 1.2.1 (Zero Object, Pointed ∞ -Category). Let C be an ∞ -category, a *zero object* is an object which is both initial and terminal. We will say that C is *pointed* if it admits a zero object. In this case, we will denote the zero object by $0 \in C$.

Obviously, if C is pointed, then the zero object is unique up to a contractible space of choices. The condition is equals to:

Proposition 1.2.1. Let C be an ∞ -category, C is pointed iff:

1. C admits initial and terminal objects $0, 1$.
2. Exists $0 \rightarrow 1$.

1.2.2 Triangle, Fiber and Cofiber

Definition 1.2.2 (Triangle, (co)Fiber Sequence). Let C be a pointed ∞ -category, a *triangle* in C is a diagram $\Delta^1 \times \Delta^1 \rightarrow C$ depicted as

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & z \end{array}$$

where 0 is the zero object. We say that a triangle in C is a *fiber sequence* if it is a pullback square, and a *cofiber sequence* if it is a pushout square.

A triangle is basically a composite of two maps

$$x \xrightarrow{f} y \xrightarrow{g} z, \quad gf \simeq 0$$

if this triangle is a pullback, then $x \simeq y \times_z 0$, where $y \times_z 0$ is exactly the fiber of g , hence $x \simeq \text{fib}(g)$. Similarly, if this triangle is a pushout, then $z \simeq y \sqcup_x 0$, where $y \sqcup_x 0$ is exactly the cofiber of f , hence $z \simeq \text{cofib}(f)$.

Remark. Let C be a pointed ∞ -category, a triangle in C consists of the following data:

1. A pair of morphisms $f : x \rightarrow y$ and $g : y \rightarrow z$ in C .
2. A 2-simplex in C corresponding to a diagram

$$\begin{array}{ccc} & y & \\ f \nearrow & & \searrow g \\ x & \xrightarrow{h} & z \end{array}$$

in C , which witnesses h as a composite of f and g .

3. A 2-simplex in C corresponding to a diagram

$$\begin{array}{ccc} & 0 & \\ 0 \nearrow & & \searrow 0 \\ x & \xrightarrow{h} & z \end{array}$$

in C , which may view as a nullhomotopy of h .

By the notion of triangle, we can define the notion of fiber and cofiber in a pointed ∞ -category.

Definition 1.2.3 ((co)Fiber). Let C be a pointed ∞ -category containing a morphism $g : x \rightarrow y$, a *fiber* of g is a fiber sequence

$$\begin{array}{ccc} w & \longrightarrow & x \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & y \end{array}$$

Dually, a *cofiber* of g is a cofiber sequence

$$\begin{array}{ccc} x & \xrightarrow{g} & y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & z \end{array}$$

We write $w = \text{fib}(g)$ and $z = \text{cofib}(g)$.

Proposition 1.2.2. Fix a pointed ∞ -category C and a morphism $f : x \rightarrow y$, if a cofiber of f exists, then it is unique up to an equivalence. Precisely, consider the full subcategory

$$E \subset \mathbf{Fun}(\Delta^1 \times \Delta^1, C)$$

spanned by the cofiber sequences, let $\theta : E \rightarrow \mathbf{Fun}(\Delta^1, C)$ be the forgetful functor, which send a diagram

$$\begin{array}{ccc} x & \xrightarrow{g} & y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & z \end{array}$$

to the morphism $g : x \rightarrow y$. Then θ is a Kan fibration. If every morphism in C admits a cofiber, then θ is a trivial Kan fibration.

Proposition 1.2.3. The functor

$$\text{cofib} : \mathbf{Fun}(\Delta^1, C) \rightarrow C$$

is the left adjoint of the following functor:

$$C \simeq \mathbf{Fun}(\Delta^0, C) \rightarrow \mathbf{Fun}(\Delta^1, C)$$

by sending every object x to the morphism $0 \rightarrow x$. Hence cofib preserves colimits.

1.2.3 Stability

Definition 1.2.4 (Stable ∞ -Category). An ∞ -category C is *stable* if it satisfies the following conditions:

1. C is pointed.
2. Every morphism in C admits a fiber and a cofiber.
3. A triangle in C is a fiber sequence if and only if it is a cofiber sequence.

Proposition 1.2.4. If C is a stable ∞ -category, then C^{op} is also stable.

1.2.4 Closure Properties

Proposition 1.2.5. Let C be a stable ∞ -category and k a simplicial set. Then the ∞ -category $\mathbf{Fun}(k, C)$ is stable.

Proposition 1.2.6. Let C be a small stable ∞ -category, and let κ be a regular cardinal, then the ∞ -category $\text{Ind}_{\kappa}(C)$ is stable.

1.3 Homotopy Category of a Stable ∞ -Category

1.3.1 Intro

Let m be a module over a commutative ring R . It admits a resolution

$$\dots \rightarrow p_2 \rightarrow p_1 \rightarrow p_0 \rightarrow m \rightarrow 0$$

by projective R -modules p_i . There are generally many different choices, but they are always *quasi-isomorphic* to each other: that is, a chain map between two resolutions that induces an isomorphism on homology. It's often convenient to work in the *derived category* $D(R)$ of the ring R , the category obtained from the category of chain complexes of R -modules by formally inverting quasi-isomorphisms.

The derived category $D(R)$ of a commutative ring R is usually not an abelian category. A morphism $f : x' \rightarrow x$ in $D(R)$ usually does not have a kernel in $D(R)$. Instead, one can associate to f its cofiber (*mapping cone*) x'' , which is well-defined up to noncanonical isomorphism.

Verdier introduced the notion of a *triangulated category* to axiomatize the properties of the derived category by forming the mapping cones, we will soon review the theory of triangulated categories and then show that the homotopy category of a stable ∞ -category is triangulated.

1.3.2 Basic Homological Algebra

Definition 1.3.5 (Additive Category, Abelian Category). Let A be a category, we say that A is *additive* if it satisfies the following conditions:

1. A admits a zero object 0 .
2. A admits finite products and coproducts.

For every pair of objects $x, y \in A$, a *zero morphism* is a map factors through $x \rightarrow 0 \rightarrow y$. It is unique, and denoted by 0 .

3. For every pair x, y the map $x \sqcup y \rightarrow x \times y$ described by

$$\begin{pmatrix} \text{id}_x & 0 \\ 0 & \text{id}_y \end{pmatrix}$$

is an isomorphism, and denote the inverse by $\varphi_{x,y}$.

We can now define the *sum* of two morphisms $f, g : x \rightarrow y$ to be the morphism $f + g : x \rightarrow y$ by the composition

$$x \xrightarrow{\Delta_x} x \times x \xrightarrow{f \times g} y \times y \xrightarrow{\varphi_{y,y}} y \sqcup y \xrightarrow{\nabla_y} y$$

where Δ_x is the diagonal map and ∇_y is the codiagonal map. This construction makes $\text{Hom}_A(x, y)$ into a commutative monoid with identity 0.

4. For every x, y the commutative monoid $\text{Hom}_A(x, y)$ is an abelian group. i.e. every morphism $f : x \rightarrow y$ admits an additive inverse $-f : x \rightarrow y$ such that $f + (-f) = 0$.

An additive category A is *abelian*, if every morphism admits a kernel and a cokernel, and every monomorphism is a kernel of its cokernel, and the canonical map

$$\text{coker}(\ker(f) \rightarrow x) \rightarrow \ker(y \rightarrow \text{coker}(f))$$

is an isomorphism for every morphism $f : x \rightarrow y$ in A .

Definition 1.3.6 (Triangulated Category). A *triangulated category* consists of the following data:

1. An additive category A .
2. A translation functor $A \rightarrow A$ which is an equivalence of categories, by $x \mapsto x[1]$.
3. A collection of *distinguished triangles*

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

These data are required to satisfy the following axioms:

1. Every morphism $f : x \rightarrow y$ in A can be extended to distinguished triangle in A .
2. The collection of distinguished triangles is stable under isomorphisms.
3. Given an object $x \in A$, the diagram

$$x \xrightarrow{\text{id}_x} x \rightarrow 0 \rightarrow x[1]$$

is a distinguished triangle.

4. A diagram

$$x \xrightarrow{f} y \xrightarrow{g} z \xrightarrow{h} x[1]$$

is a distinguished triangle if and only if the diagram

$$y \xrightarrow{g} z \xrightarrow{h} x[1] \xrightarrow{-f[1]} y[1]$$

is a distinguished triangle.

5. Given a commutative diagram

$$\begin{array}{ccccccc}
 x & \longrightarrow & y & \longrightarrow & z & \longrightarrow & x[1] \\
 f \downarrow & & \downarrow & & \downarrow & & \downarrow f[1] \\
 x' & \longrightarrow & y' & \longrightarrow & z' & \longrightarrow & x'[1]
 \end{array}$$

where the rows are distinguished triangles, then there exists a morphism $z \rightarrow z'$ making the whole diagram commutative.

6. (*Octahedral Axiom*) suppose given three distinguished triangles

$$x \xrightarrow{f} y \xrightarrow{u} y/x \xrightarrow{d} x[1], \quad y \xrightarrow{g} z \xrightarrow{v} z/y \xrightarrow{d'} y[1], \quad x \xrightarrow{g \circ f} z \xrightarrow{w} z/x \xrightarrow{d''} x[1]$$

in A , there exists a fourth distinguished triangle

$$y/x \xrightarrow{\varphi} z/x \xrightarrow{\psi} z/y \xrightarrow{\theta} y/x[1]$$

such that the diagram

commutes.

1.3.3 Homotopy Category of a Stable ∞ -Category

Let C be a stable ∞ -category. The entire construction rests on two facts: pushout squares are pullback squares, and the space of choices of a limit or colimit is contractible. The first produces the usual homological operations; the second makes them coherent.

1.3.3.1 Suspension and Additivity

Definition 1.3.7 (Suspension and Loop Objects). For $x \in C$, define Σx and Ωx by a pushout square and a pullback square, respectively:

$$\begin{array}{ccc}
 x & \longrightarrow & 0 \\
 \downarrow & & \downarrow \\
 0 & \longrightarrow & \Sigma x
 \end{array}
 \qquad
 \begin{array}{ccc}
 \Omega x & \longrightarrow & 0 \\
 \downarrow & & \downarrow \\
 0 & \longrightarrow & x
 \end{array}$$

The left square is a pushout and the right square is a pullback. We write $x[1] = \Sigma x$ and $x[-1] = \Omega x$

Proposition 1.3.7 (Suspension Is an Equivalence). The functors $\Sigma, \Omega : C \rightarrow C$ are inverse equivalences. Hence $[1] : hC \rightarrow hC$ is an equivalence.

Proof. The square defining Σx is also a pullback, so $\Omega \Sigma x \simeq x$. Dually, $\Sigma \Omega x \simeq x$. These equivalences are natural because the relevant spaces of choices are contractible. $\square$

Proposition 1.3.8 (Additivity of the Homotopy Category). The homotopy category hC is additive.

Proof. The zero map points every $\text{Map}_C(x, y)$, and suspension gives

$$\text{Map}_C(\Sigma x, y) \simeq \Omega \text{Map}_C(x, y)$$

Therefore

$$\text{Hom}_{hC}(x, y) \simeq \pi_2 \text{Map}_C(x[-2], y)$$

The right side is an abelian group, and naturality makes composition bilinear. Thus hC is preadditive.

The zero object remains a zero object in hC . For $x, y \in C$, put

$$p := \text{cofib} \left(x[-1] \xrightarrow{0} y \right)$$

The pushout universal property and the suspension equivalence give

$$\text{Map}_C(p, t) \simeq \text{Map}_C(x, t) \times \text{Map}_C(y, t)$$

Hence p is a coproduct. A finite coproduct in a preadditive category is a biproduct: its projections are determined by the matrices $(\text{id}, 0)$ and $(0, \text{id})$. Thus hC is additive. $\square$

1.3.3.2 Distinguished Triangles

For a morphism $f : x \rightarrow y$, choose a coherent diagram in C

$$\begin{array}{ccccc} x & \xrightarrow{f} & y & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C_f & \xrightarrow{\delta_f} & \Sigma x \end{array}$$

in which both small squares are pushouts. Then $C_f \simeq \text{cofib}(f)$, the outer rectangle gives Σx , and δ_f is the connecting morphism.

Definition 1.3.8 (Distinguished Triangle in hC). A triangle in hC

$$x \xrightarrow{f} y \xrightarrow{u} z \xrightarrow{\delta} x[1]$$

is *distinguished* if it is isomorphic to a cofiber triangle

$$x \xrightarrow{f} y \rightarrow C_f \xrightarrow[\delta]{f} \Sigma x$$

arising from a coherent two-pushout diagram as above.

Restriction from coherent two-pushout diagrams to the arrow f is a trivial Kan fibration. Consequently its fibers are contractible: the triangle is independent of all choices up to isomorphism in hC , and coherent squares of arrows lift to morphisms of cofiber triangles.

Lemma 1.3.1 (The Sign from Transposing a Suspension Square). Interchanging the two zero directions in a pushout square defining Σx induces $-\text{id}_{\Sigma x}$ in hC .

Proof. Under the natural identification

$$\text{Hom}_{hC}(\Sigma x, y) \simeq \pi_1 \text{Map}_C(x, y)$$

transposition is loop reversal, hence multiplication by -1 . Yoneda gives the result. $\square$

Theorem 1.3.1 (Triangulated Homotopy Category). Let C be a stable ∞ -category. With translation $x[1] = \Sigma x$ and the distinguished triangles defined above, hC is a triangulated category.

Proof. We verify Verdier's axioms.

TR1: existence, invariance, and the identity triangle. Every arrow $f : x \rightarrow y$ extends to

$$x \xrightarrow{f} y \rightarrow C_f \xrightarrow[\delta]{f} x[1]$$

by two pushouts. Isomorphism invariance is part of the definition, and $C_{\text{id}_x} \simeq 0$ gives

$$x \xrightarrow{\text{id}_x} x \rightarrow 0 \rightarrow x[1]$$

TR2: rotation. The right pushout square says that $x[1]$ is the cofiber of $y \rightarrow C_f$. One further pushout gives

$$y \xrightarrow{u} C_f \xrightarrow[\delta]{f} x[1] \xrightarrow{-f[1]} y[1]$$

The minus sign is precisely the sign of the preceding lemma. Since shifts preserve pushouts, desuspension and repeated rotation prove the converse.

TR3: morphisms of triangles. A commutative square in hC is a coherent square in C after choosing representatives and a homotopy. The trivial Kan fibration of cofiber extensions lifts it to a morphism of the two cofiber triangles. Its third component $C_f \rightarrow C_{f'}$ is the required map; uniqueness is neither asserted nor needed.

TR4: the octahedral axiom. For $x \xrightarrow{f} y \xrightarrow{g} z$, choose coherent cofiber models. Pushout pasting gives

$$\begin{array}{ccccc}
x & \xrightarrow{f} & y & \xrightarrow{g} & z \\
\downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & C_f & \xrightarrow{\varphi} & C_{gf}
\end{array}$$

The left square and outer rectangle are pushouts, hence so is the right square. Pushing it out along $C_f \rightarrow 0$ identifies its cofiber with $C_g = z \sqcup_y 0$ and produces the fourth distinguished triangle

$$C_f \xrightarrow{\varphi} C_{gf} \xrightarrow{\psi} C_g \xrightarrow{\theta} C_{f[1]}$$

All four triangles are faces of one coherent pushout diagram. Its remaining faces give exactly the commutative octahedron, including the shifted signs fixed by TR2. Thus TR4 holds and hC is triangulated. $\square$

Corollary 1.3.1 (The Stable–Triangulated Dictionary). Passing from C to hC carries the basic stable constructions to the usual triangulated ones:

stable ∞ -category	triangulated category
zero object	zero object
Σ, Ω	shifts $[1], [-1]$
finite product = finite coproduct	biproduct
cofiber/fiber sequence	distinguished triangle
exact functor	triangulated functor

Thus every formal theorem about triangulated categories applies to hC . Exact functors descend to triangulated functors, and mapping spaces give the usual long exact sequences. The enhancement C retains more: its cofibers and comparison maps are coherently functorial, unlike cones in hC .

1.4 Exact Functors

1.4.1 The Definition

Definition 1.4.9 (Exact Functor). A functor $F : C \rightarrow D$ between stable ∞ -categories is *exact* if it preserves zero objects and sends cofiber sequences in C to cofiber sequences in D . Equivalently, F preserves fiber sequences.

We have the following equivalent characterizations of exact functors.

Proposition 1.4.9. Let $F : C \rightarrow D$ be a functor between stable ∞ -categories, then the following are equivalent:

1. F is exact.

2. F preserves finite limits.
3. F preserves finite colimits.

Proof sketch. Suppose first that F is exact. It preserves suspensions, desuspensions, and finite biproducts. Moreover, the pushout of a span $y \xleftarrow{f} x \xrightarrow{g} z$ is the cofiber of

$$x \xrightarrow{(f, -g)} y + z$$

Hence F preserves pushouts. Since it also preserves the zero object, it preserves all finite colimits. Dually, or by expressing a pullback as the fiber of $y + z \rightarrow w$, it preserves all finite limits.

Conversely, a functor preserving finite colimits preserves the initial object and pushout squares, hence the zero object and cofiber sequences. A functor preserving finite limits preserves the terminal object and pullback squares; in a stable ∞ -category these are again the zero object and cofiber sequences. Thus either condition implies exactness. $\square$

1.4.2 Ext Functor

Definition 1.4.10 (Ext Functor). We write

$$\mathrm{Ext}_C^n(x, y) := \mathrm{Hom}_{hC}(x, y[n])$$

2 Homological Algebra Reviewed

2.1 t -Structures

2.1.1 Definition of t -Structures on Triangulated Categories

Recall that we say a full sub ∞ -category $C' \subset C$ is a *localization* if the inclusion functor has a left adjoint. We will introduce a class of localizations called t -structures which are particularly important in homological algebra.

Definition 2.1.1 (t -Structure on Triangulated Category). Let D be a triangulated category. A t -structure on D is a pair of full subcategories $(D_{\geq 0}, D_{\leq 0})$ stable under isomorphisms, such that the following conditions hold:

1. For $x \in D_{\geq 0}$ and $y \in D_{\leq 0}$, we have $\mathrm{Hom}_D(x, y[-1]) = 0$.
2. We have inclusions $D_{\geq 0}[1] \subset D_{\geq 0}$ and $D_{\leq 0}[-1] \subset D_{\leq 0}$.
3. For any $x \in D$, there exists a fiber sequence $x' \rightarrow x \rightarrow x''$ (i.e. $x' = \mathrm{fib}(x \rightarrow x'')$) where $x' \in D_{\geq 0}$ and $x'' \in D_{\leq -1}$.

Remark. We denote $D_{\geq n}$ for $D_{\geq 0}[n]$ and $D_{\leq n}$ for $D_{\leq 0}[n]$ for any integer n .

Remark. In this definition, either of the full subcategories $D_{\geq 0}$ and $D_{\leq 0}$ determines the other. For example, an object $y \in D$ belongs to $D_{\leq -1}$ if and only if $\mathrm{Hom}_D(x, y)$ vanishes for all $x \in D_{\geq 0}$.

2.1.2 Definition of t -Structures on Stable ∞ -Categories

Definition 2.1.2 (t -Structure on Stable ∞ -Category). Let C be a stable ∞ -category, a t -structure on C is a t -structure hC , if C is equipped with a t -structure, we let $C_{\geq n}$ and $C_{\leq n}$ denote the full subcategories of C spanned by those objects which belong to $(hC)_{\geq n}$ and $(hC)_{\leq n}$, respectively.

Proposition 2.1.1. Let C be a stable ∞ -category with a t -structure, for each $n \in \mathbb{Z}$, the full subcategory $C_{\leq n}$ is a localization of C .

Proof. Since shifts are equivalences and carry $C_{\leq n}$ to translates of $C_{\leq -1}$, it is enough to treat $n = -1$.

Recall the criterion for a full subcategory to be a localization. The inclusion $i : D \rightarrow C$ has a left adjoint if, for every $x \in C$, there is a map $\eta_x : x \rightarrow Lx$ with $Lx \in D$ such that composition with η_x induces an equivalence

$$\mathrm{Map}_C(Lx, y) \rightarrow \mathrm{Map}_C(x, y)$$

for every $y \in D$. Indeed, this is exactly the universal property characterizing Lx as the reflection of x into D ; these universal objects assemble into a functor $L : C \rightarrow D$ left adjoint to i .

Apply the third axiom of the t -structure to x . It gives a fiber, hence also cofiber, sequence

$$x' \rightarrow x \xrightarrow[x]{\eta} x''$$

with $x' \in C_{\geq 0}$ and $x'' \in C_{\leq -1}$. We claim that x'' has the required universal property. Fix $y \in C_{\leq -1}$. Applying $\mathrm{Map}_C(-, y)$ gives a fiber sequence

$$\mathrm{Map}_C(x'', y) \rightarrow \mathrm{Map}_C(x, y) \rightarrow \mathrm{Map}_C(x', y)$$

It therefore remains to show that $\mathrm{Map}_C(x', y)$ is contractible.

For $k \in \mathbb{Z}$, write

$$\mathrm{Ext}_C^k(a, b) := \mathrm{Hom}_{hC}(a, b[k]) \simeq \mathrm{Hom}_{hC}(a[-k], b)$$

If $k \leq 0$, then $-k \geq 0$, so the shift axiom gives $x'[-k] \in C_{\geq 0}$. The orthogonality axiom and $y \in C_{\leq -1}$ now imply

$$\mathrm{Ext}_C^k(x', y) \simeq \mathrm{Hom}_{hC}(x'[-k], y) \simeq 0$$

For every $m \geq 0$, stability identifies the homotopy groups of the mapping space with these groups:

$$\pi_m \mathrm{Map}_C(x', y) \simeq \mathrm{Ext}_C^{-m}(x', y) \simeq 0$$

Mapping spaces are Kan complexes. The case $m = 0$ says that this mapping space is connected, and the higher cases say that all its homotopy groups vanish; the Whitehead criterion therefore makes it contractible. Hence $\mathrm{Map}_C(x'', y) \rightarrow \mathrm{Map}_C(x, y)$ is an equivalence for every $y \in C_{\leq -1}$, so $x \mapsto x''$ defines the left adjoint $\tau_{\leq -1} : C \rightarrow C_{\leq -1}$. Shifting gives the localization $\tau_{\leq n} : C \rightarrow C_{\leq n}$ for every $n \in \mathbb{Z}$. $\square$

Corollary 2.1.1. Each $C_{\leq n}$ is closed under all limits which exist in C . Dually, each $C_{\geq n}$ is closed under all colimits which exist in C .

2.2 Truncation Functors

2.2.1 Definition of Truncation Functors

Indeed, local objects are characterized by equivalences of mapping spaces, and such equivalences are preserved under limits. The second statement is dual.

Remark (Notation). Write

$$\tau_{\leq n} : C \rightarrow C_{\leq n}, \quad \tau_{\geq n} : C \rightarrow C_{\geq n}$$

for the left and right adjoints to the respective inclusions. Every $x \in C$ then fits naturally into a fiber-cofiber sequence

$$\tau_{\geq n}x \rightarrow x \rightarrow \tau_{\leq n-1}x$$

Both truncation functors preserve every $C_{\leq m}$ and every $C_{\geq m}$. For example, $\tau_{\leq n}$ either acts as the identity on $C_{\leq m}$ or lands in the smaller subcategory $C_{\leq n}$. The assertion for $\tau_{\geq n}$ follows from the displayed fiber sequence and closure under limits. The remaining statements are dual.

Remark (Warning). The symbol $\tau_{\leq n}$ here denotes truncation for a t -structure, not the truncation of an arbitrary ∞ -category by homotopy dimension. A nonzero object of a stable ∞ -category is never truncated in the latter sense: for arbitrarily large r , its identity gives a nonzero class in $\pi_r \text{Map}_C(x[-r], x)$.

The notions agree after restricting the source to $C_{\geq 0}$. For $k \geq -1$, an object x belongs to $C_{\leq k}$ exactly when $\text{Map}_C(y, x)$ is k -truncated for every $y \in C_{\geq 0}$.

2.2.2 Commuting Truncations and the Heart

For $m, n \in \mathbb{Z}$, the universal properties give a natural transformation

$$\theta : \tau_{\leq m} \circ \tau_{\geq n} \rightarrow \tau_{\geq n} \circ \tau_{\leq m}$$

Proposition 2.2.2. The transformation θ is an equivalence. If $m < n$, both sides are zero; if $m \geq n$, they define the truncation of x to the interval $[n, m]$.

Proof sketch. Both sides lie in $C_{\geq n} \cap C_{\leq m}$. When $m < n$, orthogonality makes this intersection zero. When $m \geq n$, apply $\text{Ext}_C^0(-, y)$ for $y \in C_{\geq n} \cap C_{\leq m}$ to the two truncation sequences. Orthogonality identifies the outer maps, and the five lemma identifies the middle map induced by θ . Yoneda then shows that θ is an equivalence. $\square$

Definition 2.2.3 (Heart). The *heart* of the t -structure is

$$C^\heartsuit := C_{\geq 0} \cap C_{\leq 0}$$

Define the homology-object functors by

$$\pi_0 := \tau_{\leq 0} \circ \tau_{\geq 0} \simeq \tau_{\geq 0} \circ \tau_{\leq 0}, \quad \pi_n(x) := \pi_0(x[-n])$$

For $x, y \in C^\heartsuit$ and $r > 0$, orthogonality gives

$$\pi_r \text{Map}_C(x, y) \simeq \text{Ext}_C^{-r}(x, y) \simeq 0$$

Thus mapping spaces in $C^\heartsuit$ are discrete, so this ∞ -category is the nerve of its homotopy category. The category $hC^\heartsuit$ is abelian; kernels and cokernels are obtained by applying π_0 to fibers and cofibers. Our indexing is homological: the object above is denoted $\pi_n(x)$ rather than $H_n(x)$.

2.2.3 General Localizations

Not every localization of a stable ∞ -category comes from a t -structure. Given a collection S of morphisms, the localization $S^{-1}C$ is the full subcategory of S -local objects. An object x is S -local when, for every $f : y' \rightarrow y$ in S , composition with f induces an equivalence

$$\mathrm{Map}_C(y, x) \rightarrow \mathrm{Map}_C(y', x)$$

To compare this with a t -structure, complete f to a cofiber sequence

$$y' \rightarrow y \rightarrow y''$$

Applying $\mathrm{Ext}_C^i(-, x)$ gives

$$\dots \rightarrow \mathrm{Ext}_C^i(y'', x) \rightarrow \mathrm{Ext}_C^i(y, x) \rightarrow \mathrm{Ext}_C^i(y', x) \rightarrow \mathrm{Ext}_C^{i+1}(y'', x) \rightarrow \dots$$

Consequently, x is f -local exactly when

$$\mathrm{Ext}_C^i(y, x) \rightarrow \mathrm{Ext}_C^i(y', x)$$

is an isomorphism for every $i \leq 0$. In particular, locality implies $\mathrm{Ext}_C^i(y'', x) = 0$ for $i \leq 0$, while vanishing for $i \leq 1$ is sufficient for locality. If y' is zero, the exact criterion reduces simply to $\mathrm{Ext}_C^i(y, x) = 0$ for every $i \leq 0$.

2.3 Some More Propositions

2.3.1 Quasisaturated Localizations

Definition 2.3.4 (Quasisaturated Collection). Let C admit pushouts. A collection S of morphisms is *quasisaturated* if

1. every equivalence belongs to S ;
2. S satisfies two-out-of-three for every composable pair;
3. a pushout of a morphism in S again belongs to S .

Intersections of quasisaturated collections are quasisaturated. Hence every collection of maps has a smallest quasisaturated closure, called the collection it *generates*.

Definition 2.3.5 (Closed under Extensions). A full subcategory $E \subset C$ is *closed under extensions* if, for every fiber-cofiber sequence $x \rightarrow y \rightarrow z$, the conditions $x, z \in E$ imply $y \in E$.

Let $L : C \rightarrow C$ be an idempotent localization, let C_L be its essential image, and let S_L be the collection of maps f for which $L(f)$ is an equivalence. The collection S_L is quasisaturated.

Proposition 2.3.3 (t -Structure Localizations). The following conditions are equivalent:

1. S_L is generated by a collection of maps of the form $0 \rightarrow a$;
2. S_L is generated by all $0 \rightarrow a$ such that $La \simeq 0$;
3. the local subcategory C_L is closed under extensions;
4. for $a \in C$ and $b \in C_L$, the map $\mathrm{Ext}_C^1(La, b) \rightarrow \mathrm{Ext}_C^1(a, b)$ is injective;
5. $C_{\geq 0} := \{a \mid La \simeq 0\}$ and $C_{\leq -1} := \{a \mid La \simeq a\}$ determine a t -structure on C .

Proof sketch. The implication (1)→(2) follows by enlarging the chosen generators to all L -acyclic objects. For (2)→(3), mapping an acyclic generator into a fiber sequence shows that an extension of local objects is local.

If (3) holds, an extension of La by a local object has local middle term; consequently an extension class which becomes split after pullback along $a \rightarrow La$ was already split. This is (4).

Under (4), the fiber sequence

$$\mathrm{fib}(a \rightarrow La) \rightarrow a \rightarrow La$$

gives the required truncation sequence. Adjunction gives orthogonality, and the injectivity in (4) supplies the remaining shifted orthogonality. Hence the two classes in (5) form a t -structure. Finally, if (5) holds, the cofiber of every map inverted by L is in $C_{\geq 0}$; pushouts and two-out-of-three show that the maps $0 \rightarrow a$ with $a \in C_{\geq 0}$ generate all of S_L . $\square$

Thus the localizations arising as negative truncations are exactly those generated, up to quasi-saturation, by killing objects.

2.3.2 Boundedness and Completion

Definition 2.3.6 (Bounded Objects). Let C^+ consist of the objects lying in some $C_{\geq n}$, and let C^- consist of those lying in some $C_{\leq n}$. Put

$$C^b := C^+ \cap C^-$$

The t -structure is *left bounded*, *right bounded*, or *bounded* when $C = C^+$, $C = C^-$, or $C = C^b$, respectively.

Definition 2.3.7 (Left Completion). The *left completion* $\hat{C}$ is the ∞ -category of compatible Postnikov towers $(x_n)_{n \in \mathbb{Z}}$ with

$$x_n \in C_{\leq n}, \quad \tau_{\leq n} x_{n+1} \simeq x_n$$

Equivalently,

$$\hat{C} \simeq \lim_n C_{\leq n}$$

The canonical functor is

$$C \rightarrow \hat{C}, \quad x \mapsto (\tau_{\leq n} x)_n$$

We call C *left complete* when this functor is an equivalence.

Proposition 2.3.4 (Basic Properties of Completion). The left completion has the following properties:

1. $\hat{C}$ is stable and inherits a t -structure;
2. the functor $C \rightarrow \hat{C}$ is exact and induces equivalences

$$C_{\leq n} \simeq \hat{C}_{\leq n}$$

for every n ;

3. it induces an equivalence $C^+ \simeq \hat{C}^+$ on left bounded objects.

Proof sketch. Limits and cofibers of compatible towers are formed levelwise, with the shift reindexing the tower, so $\hat{C}$ is stable. The two halves of its t -structure are detected levelwise. A tower bounded above stabilizes at the corresponding truncation, which proves (2); shifting proves (3). $\square$

In particular, left completion changes only objects which are unbounded below. Passing to the left bounded part and taking left completion are inverse constructions. Right completion and right completeness are defined dually.

Proposition 2.3.5 (Criterion for Left Completeness). Suppose that C admits countable products and that $C_{\geq 0}$ is closed under them. Then the following are equivalent:

1. C is left complete;
2. the only object in $\bigcap_{n \in \mathbb{Z}} C_{\geq n}$ is the zero object.

Proof sketch. Every countable tower has a limit, computed by the fiber sequence

$$\lim_n x_n \rightarrow \prod_n x_n \xrightarrow{\text{id} - \text{shift}} \prod_n x_n$$

Hence a compatible Postnikov tower has a limit in C . For every x , the cofiber of the completion map

$$x \rightarrow \lim_n \tau_{\leq n} x$$

belongs to every $C_{\geq n}$. Condition (2) therefore makes this map an equivalence. Conversely, if x belongs to every $C_{\geq n}$, then all its truncations vanish; left completeness forces $x \simeq 0$. $\square$

2.4 Filtered Objects and Spectral Sequences

Let C be a stable ∞ -category with a t -structure and heart $A \simeq C^\heartsuit$. A *filtered object* is a functor $x : N(\mathbb{Z}) \rightarrow C$, depicted as

$$\dots \rightarrow x_{-1} \xrightarrow{f^0} x_0 \xrightarrow{f^1} x_1 \rightarrow \dots$$

Its p th associated graded object is

$$\text{gr}^p x := \text{cofib} (x_{p-1} \rightarrow x_p)$$

We now construct a spectral sequence in A beginning with

$$E_1^{p,q} \simeq \pi_{p+q} \text{gr}^p x$$

2.4.1 Interval Cofibers

Definition 2.4.8 (Complex Associated to a Filtration). Adjoin a least element $-\infty$ to $\mathbb{Z}$. The filtered object x extends to an interval diagram with values $x_{i,j}$ for $-\infty \leq i \leq j$, defined by

$$x_{-\infty,j} := x_j, \quad x_{i,j} := \text{cofib} (x_i \rightarrow x_j)$$

Thus $x_{i,i} \simeq 0$, and for $i \leq j \leq k$ there is a canonical fiber-cofiber sequence

$$x_{i,j} \rightarrow x_{i,k} \rightarrow x_{j,k} \rightarrow x_{i,j}[1]$$

Equivalently, the square with vertices $x_{i,j}, x_{i,k}, 0, x_{j,k}$ is a pushout. Conversely, every interval diagram satisfying these two properties is determined by the filtration $x_{-\infty,j}$. More precisely, restriction to the $-\infty$ th row is an equivalence of ∞ -categories. Hence all interval cofibers and connecting maps above are functorial and independent of choices up to a contractible space.

Remark (Underlying Chain Complex). Set

$$c_p := x_{p-1,p}[-p]$$

The connecting maps of adjacent intervals give

$$\dots \rightarrow c_1 \rightarrow c_0 \rightarrow c_{-1} \rightarrow \dots$$

in hC . Two consecutive maps compose to zero because they are adjacent boundary maps in a coherent pushout diagram. Thus every filtered object has a canonical chain complex of graded pieces in its triangulated homotopy category.

2.4.2 The Spectral Sequence

Proposition 2.4.6 (Spectral Sequence of a Filtered Object). For $r \geq 1$, define an object of the heart by

$$E_r^{p,q} := \text{im}(\pi_{p+q} x_{p-r,p} \rightarrow \pi_{p+q} x_{p-1,p+r-1})$$

The connecting maps of the interval sequences induce differentials

$$d_r : E_r^{p,q} \rightarrow E_r^{p-r,q+r-1}$$

They satisfy $d_r^2 = 0$, and there are natural isomorphisms

$$E_{r+1}^{p,q} \simeq \ker(d_r : E_r^{p,q} \rightarrow E_r^{p-r,q+r-1}) / \text{im}(d_r : E_r^{p+r,q-r+1} \rightarrow E_r^{p,q})$$

In particular, (E_r, d_r) is a spectral sequence in A , natural in the filtered object x .

Proof sketch. For $r = 1$, both terms in the defining image are $x_{p-1,p}$, giving the announced E_1 -page. For general r , the differential is induced by the boundary map between the two relevant interval cofibers. The pushout identities imply that two successive boundary maps compose to zero. Exactness in the heart then identifies the image defining E_{r+1} with the kernel of d_r modulo the image of the incoming d_r . $\square$

The differential has bidegree $(-r, r-1)$ and lowers total degree by one. On the first page one may equivalently write

$$E_1^{p,q} \simeq \pi_q c_p$$

so d_1 is simply the homology differential obtained from the underlying chain complex of graded pieces.

Remark. If C is the derived ∞ -category of an abelian category and x is a filtered chain complex, this construction recovers the classical spectral sequence of a filtered complex. The interval-diagram formulation is useful because it makes every higher differential canonical and functorial.

2.4.3 Convergence

Definition 2.4.9 (Compatibility with Sequential Colimits). The t -structure on C is *compatible with sequential colimits* if C admits colimits indexed by $\mathbb{N}$ and $C_{\leq 0}$ is closed under them. Under this assumption the functors $\pi_n : C \rightarrow A$ preserve sequential colimits, and sequential colimits in the heart are exact.

Proposition 2.4.7 (Convergence for a Bounded-Below Filtration). Suppose the t -structure is compatible with sequential colimits and $x_p \simeq 0$ for all sufficiently negative p . Put

$$x_\infty := \operatorname{colim}_p x_p, \quad a_n := \pi_n x_\infty$$

Then the spectral sequence converges to $\pi_* x_\infty$. Explicitly, for fixed (p, q) the differentials eventually vanish, and a_n has the exhaustive increasing filtration

$$F^p a_n := \operatorname{im}(\pi_n x_p \rightarrow a_n)$$

whose associated graded pieces are

$$E_\infty^{p,q} \simeq F^p a_{p+q} / F^{p-1} a_{p+q}$$

Proof sketch. The lower bound makes the incoming and outgoing intervals defining d_r eventually constant or zero, so every bidegree stabilizes. Since π_n commutes with the sequential colimit, the images of $\pi_n x_p \rightarrow \pi_n x_\infty$ form an exhaustive filtration. Applying the long exact sequence to

$$x_{p-1} \rightarrow x_p \rightarrow \operatorname{gr}^p x$$

identifies the stable page with the indicated subquotient. $\square$

2.5 The Dold–Kan Correspondence

The classical Dold–Kan correspondence says that a simplicial object in an abelian category is exactly a chain complex concentrated in nonnegative degrees. Its stable ∞ -categorical form replaces chain complexes by nonnegative filtrations; this retains all higher coherences and connects directly with the preceding spectral sequence.

2.5.1 Normalization and Denormalization

Let A be an abelian category and let $x : \Delta^{\operatorname{op}} \rightarrow A$ be a simplicial object, with face maps d_i and degeneracy maps s_i . Its *normalized chain complex* Nx is defined by

$$N_n x := \bigcap_{i=1}^n \ker(d_i : x_n \rightarrow x_{n-1}), \quad \partial := d_0 : N_n x \rightarrow N_{n-1} x$$

The simplicial identities give $\partial^2 = 0$. If

$$D_n x := \sum_{i=0}^{n-1} \text{im}(s_i)$$

is the degenerate subobject, then there is a canonical decomposition

$$x_n \simeq N_n x \oplus D_n x$$

More generally, in an additive category the normalized piece is the image of the idempotent

$$p_n := (1 - s_{n-1}d_n) \dots (1 - s_0d_1)$$

whenever this idempotent splits.

Definition 2.5.10 (The Dold–Kan Functor). Let $c = (c_n, \partial)$ be a chain complex in nonnegative degrees. The *Dold–Kan functor* constructs a simplicial object $\text{DK}(c)$ whose degree- n object is

$$\text{DK}(c)_n \simeq \bigoplus_{\eta: [n] \twoheadrightarrow [k]} c_k$$

where the sum runs over all order-preserving surjections $\eta: [n] \twoheadrightarrow [k]$. For $\alpha: [m] \rightarrow [n]$, factor

$$\eta \circ \alpha: [m] \twoheadrightarrow [\ell] \hookrightarrow [k]$$

into a surjection followed by an injection. On the η -summand, the induced structure map is the identity if the injection is the identity, is $\partial: c_k \rightarrow c_{k-1}$ if it omits only 0, and is zero otherwise. This rule defines all faces and degeneracies; the equality $\partial^2 = 0$ gives the simplicial identities.

Thus the copies indexed by nonidentity surjections freely supply exactly the degenerate simplices missing from a chain complex. For example, a complex concentrated in degree zero gives a constant simplicial object, while a complex with a in degree one and zero differential has $\text{DK}(a[1])_n \simeq a^{\oplus n}$.

Theorem 2.5.1 (Classical Dold–Kan). For every additive category A , the functor

$$\text{DK}: \text{Ch}_{\geq 0}(A) \rightarrow \mathbf{Fun}(\Delta^{\text{op}}, A)$$

is fully faithful. It is an equivalence when A is idempotent complete. In particular, for every abelian category there are inverse equivalences

$$\text{DK}: \text{Ch}_{\geq 0}(A) \xrightarrow{\sim} \mathbf{Fun}(\Delta^{\text{op}}, A) : N$$

Proof sketch. The simplicial identities produce a normalization idempotent on x_n . Its image is $N_n x$, and splitting the complementary degeneracy idempotents yields the canonical decomposition

$$x_n \simeq \bigoplus_{\eta: [n] \twoheadrightarrow [k]} N_k x$$

This identifies $N \text{DK}(c)$ with c and $\text{DK}(Nx)$ with x . Additivity is enough for full faithfulness; idempotent completeness is exactly what guarantees the required splittings and hence essen-

tial surjectivity. In an abelian category the normalization can equivalently be formed by the displayed intersections of kernels. $\square$

Corollary 2.5.2. If x is a simplicial abelian group, then its underlying simplicial set is a Kan complex and

$$\pi_n(x) \simeq H_n(Nx)$$

Thus simplicial homotopy groups become ordinary homology groups under normalization.

2.5.2 Stable Dold–Kan

For a stable ∞ -category C , write

$$\mathrm{Fil}_{\geq 0}(C) := \mathbf{Fun}(N_{\bullet}(\mathbb{N}), C), \quad sC := \mathbf{Fun}(N_{\bullet}(\Delta)^{\mathrm{op}}, C)$$

Objects of the first category are sequences $d_0 \rightarrow d_1 \rightarrow \dots$, while objects of the second are simplicial objects of C .

Theorem 2.5.2 (Stable Dold–Kan). Every stable ∞ -category C admits inverse equivalences

$$\mathrm{DK} : \mathrm{Fil}_{\geq 0}(C) \xrightarrow{\sim} sC : N^{\mathrm{st}}$$

No t -structure, countable colimits, or idempotent-completeness assumption is required.

The stable normalization N^{st} sends a simplicial object x to its skeletal filtration

$$d_n := |\mathrm{sk}_n x|$$

where $|\mathrm{sk}_n x|$ is the finite geometric realization of the n -skeleton. If $\ell_n x$ denotes the latching object, stability gives

$$\mathrm{cofib}(d_{n-1} \rightarrow d_n)[-n] \simeq \mathrm{cofib}(\ell_n x \rightarrow x_n)$$

The right-hand side is the stable normalized object in degree n . Hence the underlying chain complex of this filtration is the normalized chain complex of x in hC . The equivalence itself contains the higher null-homotopies and compatibilities which that chain complex alone forgets.

Remark (Relation with the Spectral Sequence). Suppose C has a t -structure with heart A . Applying π_q degree-wise to x gives a simplicial object of A , and the first page of the spectral sequence of its skeletal filtration is

$$E_1^{p,q} \simeq N_p(\pi_q x)$$

Its first differential is the normalized differential. If the geometric realization exists and the convergence hypotheses above hold, then

$$E_1^{p,q} \implies \pi_{p+q}|x|$$

In this sense the stable Dold–Kan equivalence is the structural source of the spectral sequence attached to a simplicial object.

3 Derived ∞ -Categories

3.1 Nerves of dg-Categories

3.1.1 Introduction of the Goal

Let A be an additive category. The chain complexes in A can be organized into an ∞ -category C , described informally as follows:

1. The objects of C are chain complexes in A .
2. Given $x_*, y_* \in C$, the morphisms $x_* \rightarrow y_*$ are chain maps.
3. Given $f, g : x_* \rightarrow y_*$, a 2-morphism $f \rightarrow g$ is represented by a chain homotopy, that is, maps $h_n : x_n \rightarrow y_{n+1}$ satisfying

$$d_y \circ h_n + h_{n-1} \circ d_x = f_n - g_n$$

4. ...

To make this precise, we could proceed in several steps:

1. Let $\mathbf{Ch}(A)$ be the ordinary category of chain complexes in A .
2. For $x_*, y_* \in \mathbf{Ch}(A)$, form the mapping chain complex

$$\underline{\mathrm{Hom}}_A(x_*, y_*)_m := \prod_{n \in \mathbb{Z}} \mathrm{Hom}_A(x_n, y_{n+m})$$

with differential

$$(\partial f)_n := d_y f_n - (-1)^m f_{n-1} d_x$$

Its degree-zero cycles are chain maps, its degree-zero boundaries are null-homotopic maps, and

$$H_m(\underline{\mathrm{Hom}}_A(x_*, y_*)) \simeq \mathrm{Hom}_{h\mathbf{Ch}(A)}(x_*, y_*[-m])$$

Composition is a chain map, so $\mathbf{Ch}(A)$ is enriched over $\mathbf{Ch}(\mathbf{Ab})$; in other words, it is a dg-category.

3. The connective truncation

$$\tau_{\geq 0} : \mathbf{Ch}(\mathbf{Ab}) \rightarrow \mathbf{Ch}_{\geq 0}(\mathbf{Ab})$$

is right-lax monoidal. Concretely, it comes with coherent maps

$$\tau_{\geq 0} m \otimes \tau_{\geq 0} n \rightarrow \tau_{\geq 0}(m \otimes n)$$

Applying it to every mapping complex preserves units and composition. Thus $\mathbf{Ch}(A)$ is enriched over nonnegatively graded chain complexes.

4. By the Dold–Kan correspondence from the preceding chapter,

$$\mathrm{DK} : \mathbf{Ch}_{\geq 0}(\mathbf{Ab}) \rightarrow \mathbf{Fun}(\Delta^{\mathrm{op}}, \mathbf{Ab})$$

is an equivalence. The Alexander–Whitney construction equips $\mathbf{DK}$ with a right-lax monoidal structure, so it again transports enrichment. The resulting simplicial abelian group of morphisms is

$$\mathrm{Map}_\Delta(x_*, y_*) := \mathbf{DK}(\tau_{\geq 0} \underline{\mathrm{Hom}}_A(x_*, y_*))$$

5. Forgetting the abelian-group structure degreewise turns these mapping objects into simplicial sets. Their bilinear composition gives a simplicial category, denoted $\mathbf{Ch}_\Delta(A)$. Every simplicial abelian group is a Kan complex, so every mapping simplicial set of $\mathbf{Ch}_\Delta(A)$ is Kan; hence this simplicial category is fibrant.
6. Finally, take the homotopy coherent nerve

$$C_{\mathrm{dg}}(A) := N_{\mathrm{hc}}(\mathbf{Ch}_\Delta(A))$$

Since $\mathbf{Ch}_\Delta(A)$ is fibrant, $C_{\mathrm{dg}}(A)$ is an ∞ -category. It has the same objects as $\mathbf{Ch}(A)$, while its mapping spaces retain chain maps, chain homotopies, and all higher coherent homotopies.

This procedure is indeed too complicated, so we will introduce some new concepts to simplify it.

3.1.2 What is a dg(differential graded)-category?

Definition 3.1.1 (dg-Category). Let k be a commutative ring. A *dg-category* over k consists of the following data:

1. A collection $\{x, y, \dots\}$ whose elements are called *objects* of C .
2. For every pair of $x, y \in C$, a *chain complex* of k -modules

$$\dots \rightarrow \mathrm{Map}_C(x, y)_1 \rightarrow \mathrm{Map}_C(x, y)_0 \rightarrow \mathrm{Map}_C(x, y)_{-1} \rightarrow \dots$$

which we denote by $\mathrm{Map}_C(x, y)_*$.

3. For every triple $x, y, z \in C$ a composition map

$$\mathrm{Map}_C(y, z)_* \otimes_k \mathrm{Map}_C(x, y)_* \rightarrow \mathrm{Map}_C(x, z)_*$$

which we can identify with a collection of k -bilinear maps

$$\circ : \mathrm{Map}_C(y, z)_p \times \mathrm{Map}_C(x, y)_q \rightarrow \mathrm{Map}_C(x, z)_{p+q}$$

satisfying the *Leibniz rule*

$$d(g \circ f) = dg \circ f + (-1)^p g \circ df$$

4. For each $x \in C$, an *identity morphism* $\mathrm{id}_x \in \mathrm{Map}_C(x, x)_0$ such that

$$g \circ \mathrm{id}_x = g, \quad \mathrm{id}_x \circ f = f$$

for all $f \in \mathrm{Map}_C(y, x)_p$ and $g \in \mathrm{Map}_C(x, y)_q$.

The composition law is required to be associative in the following sense: for every triple $f \in \mathrm{Map}_C(w, x)_p$, $g \in \mathrm{Map}_C(x, y)_q$ and $h \in \mathrm{Map}_C(y, z)_r$, we have

$$h \circ (g \circ f) = (h \circ g) \circ f$$

In the case $k = \mathbb{Z}$, we refer to C as a *dg-category* without further qualification.

We now introduce a construction called the dg-nerve.

Definition 3.1.2 (dg-Nerve). Let C be a dg-category, we associate to C to a simplicial set $N_{\bullet}^{\text{dg}}(C)$ called the *dg-nerve*. For each $n \geq 0$, we define

$$N_{\bullet}^{\text{dg}}(C)_n \simeq \text{Hom}_{\mathbf{sSet}}(\Delta^n, N_{\bullet}^{\text{dg}}(C))$$

to be the set of all ordered pairs $(\{x_i\}_{0 \leq i \leq n}, \{f_I\})$ where:

1. For $0 \leq i \leq n$, x_i denotes an object of the dg-category C .
2. For every subset $I = \{i_- < i_m < i_{m-1} < \dots < i_1 < i_+\} \subset [n]$ with $m \geq 0$, f_I is an element of the abelian group $\text{Map}_C(x_{i_-}, x_{i_+})_m$, satisfying

$$df_I = \sum_{1 \leq r \leq m} (-1)^r (f_{I - \{i_r\}} - f_{i_r < \dots < i_1 < i_+} \circ f_{i_- < i_m < \dots < i_r})$$

If $\alpha : [m] \rightarrow [n]$ is a nondecreasing function, then the induced map $N_{\bullet}^{\text{dg}}(C)_n \rightarrow N_{\bullet}^{\text{dg}}(C)_m$ is given by

$$(\{x_i\}_{0 \leq i \leq n}, \{f_I\}) \mapsto (\{x_{\alpha(j)}\}_{0 \leq j \leq m}, \{g_J\})$$

where

$$g_J = \begin{cases} f_{\alpha(J)} & \text{if } \alpha|_J \text{ is injective} \\ \text{id}_{x_i} & \text{if } J = \{j, j'\} \text{ with } \alpha(j) = \alpha(j') = i \\ 0 & \text{otherwise} \end{cases}$$

Example. Let C be a dg-category, then:

1. A 0-simplex of $N_{\bullet}^{\text{dg}}(C)$ is an object of C .
2. A 1-simplex of $N_{\bullet}^{\text{dg}}(C)$ is a morphism of C , that is, a pair of objects $x, y \in C$ together with an element $f \in \text{Map}_C(x, y)_0$ satisfying $df = 0$.
3. A 2-simplex of $N_{\bullet}^{\text{dg}}(C)$ is a triple of objects $x, y, z \in C$ together with triple of morphisms

$$f \in \text{Map}_C(x, y)_0, \quad g \in \text{Map}_C(y, z)_0, \quad h \in \text{Map}_C(x, z)_0$$

satisfying $df = dg = dh = 0$, together with an element $k \in \text{Map}_C(x, z)_1$ satisfying $dk = (g \circ f) - h$.

Remark. Let C be a dg-category and let C_0 denote its underlying ordinary category. Then $N_{\bullet}(C_0)$ is isomorphic to the simplicial subset of $N_{\bullet}^{\text{dg}}(C)$ whose n -simplices are the pairs $(\{x_i\}_{0 \leq i \leq n}, \{f_I\})$ with $f_I = 0$ whenever I has more than two elements. In particular, $N_{\bullet}(C_0) \rightarrow N_{\bullet}^{\text{dg}}(C)$ is bijective on n -simplices for $n \leq 1$.

Proposition 3.1.1 (The dg-Nerve is an ∞ -Category). If C is a dg-category, then the simplicial set $N_{\bullet}^{\text{dg}}(C)$ is an ∞ -category.

Proof. It is enough to fill every inner horn. Fix $0 < j < n$ and a map

$$\varphi_0 : \Lambda_j^n \rightarrow N_{\bullet}^{\text{dg}}(C)$$

Unwinding the definition, the horn specifies objects $x_0, \dots, x_n$ and elements f_I satisfying the dg-nerve relation for every nonempty subset $I \subseteq [n]$, except for

$$I = [n], \quad I = [n] - \{j\}$$

Indeed, these are precisely the two nondegenerate simplices which do not lie in the j th horn. Define the missing elements by

$$f_{[n]-\{j\}} := \sum_{0 < p < n} (-1)^{p-j} f_{\{p, p+1, \dots, n\}} \circ f_{\{0, \dots, p\}} - \sum_{0 < p < n, p \neq j} (-1)^{p-j} f_{[n]-\{p\}}$$

and set

$$f_{[n]} := 0$$

We verify that these choices satisfy the two missing relations. With $f_{[n]} = 0$, the relation indexed by $[n]$ has zero left-hand side; solving its right-hand side for the only unknown term $f_{[n]-\{j\}}$ gives exactly the displayed formula.

It remains to check the relation indexed by $[n] - \{j\}$. For any subset I , let R_I be the left-hand side minus the right-hand side of its dg-nerve relation. Expanding $dR_{[n]}$ with $d^2 = 0$ and the Leibniz rule, the terms obtained by deleting two vertices cancel in pairs, and the threefold composites cancel by associativity. Since $R_I = 0$ for every proper face already contained in the horn, the result reduces to

$$dR_{[n]} = (-1)^j R_{[n]-\{j\}}$$

Our definition gives $R_{[n]} = 0$, so its differential vanishes and hence $R_{[n]-\{j\}} = 0$. We have therefore obtained an n -simplex extending φ_0 . Every inner horn admits a filler, so $N_{\bullet}^{\text{dg}}(C)$ is an ∞ -category. $\square$

Remark (The dg-Nerve Need Not Be Stable). The preceding proposition asserts only that $N_{\bullet}^{\text{dg}}(C)$ is an ∞ -category. It need not be stable. For example, let C have one object x with

$$\text{Map}_C(x, x)_* = k[0]$$

for a nonzero ring k . Then $\text{Map}_{N_{\bullet}^{\text{dg}}(C)}(x, x)$ is the discrete space underlying k , hence is not contractible. Since x is the only object, a zero object would have to be x ; therefore $N_{\bullet}^{\text{dg}}(C)$ is not even pointed and cannot be stable.

Stability requires additional structure on C . A standard sufficient hypothesis is that C be *pretriangulated*: it contains a zero object and is closed under shifts and mapping cones in a way compatible with its dg enrichment. Under this hypothesis $N_{\bullet}^{\text{dg}}(C)$ is stable. In particular, the dg-category $\mathbf{Ch}(A)$ of chain complexes is pretriangulated, which is why $\mathbf{Ch}_{\infty}(A) = N_{\bullet}^{\text{dg}}(\mathbf{Ch}(A))$ is stable in the next section.

3.1.3 Mapping Spaces and the Coherent Nerve

For a dg-category C , let hC denote the ordinary category with the same objects and morphism sets

$$\text{Hom}_{hC}(x, y) := H_0(\text{Map}_C(x, y)_*)$$

Proposition 3.1.2 (Homotopy Category and Mapping Spaces). There is a canonical isomorphism

$$hC \simeq hN_{\bullet}^{\text{dg}}(C)$$

For every $x, y \in C$, the right mapping space of the dg-nerve is naturally isomorphic to

$$\text{Map}_{N_{\bullet}^{\text{dg}}(C)}^R(x, y) \simeq \text{DK}(\tau_{\geq 0} \text{Map}_C(x, y)_*)$$

Consequently, for every $m \geq 0$,

$$\pi_m \text{Map}_{N_{\bullet}^{\text{dg}}(C)}(x, y) \simeq H_m(\text{Map}_C(x, y)_*)$$

Proof sketch. The canonical map from the nerve of the closed degree-zero morphisms to $N_{\bullet}^{\text{dg}}(C)$ is the identity on objects and morphisms. A 2-simplex shows that two cycles $f, g \in \text{Map}_C(x, y)_0$ represent the same morphism exactly when $f - g = dh$ for some $h \in \text{Map}_C(x, y)_1$. This gives the first isomorphism.

An n -simplex of the right mapping space has fixed initial and terminal vertices x, y . Unwinding the dg-nerve relation identifies its remaining elements, after the standard change of signs, with an n -simplex of the Dold–Kan object of $\tau_{\geq 0} \text{Map}_C(x, y)_*$. The last assertion is then the identity $\pi_m \text{DK}(c) \simeq H_m(c)$. $\square$

The same formula constructs a simplicial category C_{Δ} . Its objects are those of C , and

$$\text{Map}_{C_{\Delta}}(x, y) := \text{DK}(\tau_{\geq 0} \text{Map}_C(x, y)_*)$$

Composition is induced by the right-lax monoidal Alexander–Whitney map. Every mapping object is a simplicial abelian group and hence a Kan complex, so C_{Δ} is a fibrant simplicial category.

Proposition 3.1.3 (Comparison of the Two Nerves). There is a natural functor

$$\theta : N_{\text{hc}}(C_{\Delta}) \rightarrow N_{\bullet}^{\text{dg}}(C)$$

which is an equivalence of ∞ -categories.

Proof sketch. The functor is the identity on objects and is obtained by applying the Alexander–Whitney formula to coherent simplices. On every pair x, y , the preceding proposition identifies the induced map of right mapping spaces with the identity of $\text{DK}(\tau_{\geq 0} \text{Map}_C(x, y)_*)$. Thus θ is fully faithful and essentially surjective, hence an equivalence. $\square$

This comparison explains why the direct formula for $N_{\bullet}^{\text{dg}}(C)$ and the longer construction by simplicial enrichment encode the same ∞ -category.

3.1.4 Functoriality and the Model Structure

Definition 3.1.3 (dg-Functor). Let C, D be dg-categories over a commutative ring k . A dg-functor $F : C \rightarrow D$ assigns an object $F(x)$ to every $x \in C$ and a chain map

$$F_{x,y} : \text{Map}_C(x, y)_* \rightarrow \text{Map}_D(F(x), F(y))_*$$

to every pair x, y , compatibly with identities and composition. It induces functors $hF : hC \rightarrow hD$ and $N_*^{\text{dg}}(F) : N_*^{\text{dg}}(C) \rightarrow N_*^{\text{dg}}(D)$.

Theorem 3.1.1 (Model Structure on dg-Categories). The category $\mathbf{dgCat}_k$ of small dg-categories over k admits a combinatorial model structure characterized as follows.

1. A dg-functor $F : C \rightarrow D$ is a weak equivalence, also called a *quasi-equivalence*, exactly when hF is an equivalence and every map

$$\text{Map}_C(x, y)_* \rightarrow \text{Map}_D(F(x), F(y))_*$$

is a quasi-isomorphism.

2. It is a fibration exactly when these maps of complexes are degreewise surjective and hF is an isofibration: every isomorphism $F(x) \rightarrow y$ in hD lifts to an isomorphism $x \rightarrow x'$ in hC .

Proposition 3.1.4 (The dg-Nerve is Right Quillen). Let L_{dg} denote the left adjoint of the dg-nerve. There is a Quillen adjunction

$$L_{\text{dg}} : \mathbf{sSet}_{\text{Joyal}} \rightleftarrows \mathbf{Cat}_{\text{dg},k} : N_*^{\text{dg}}$$

Thus L_{dg} is left Quillen and N_*^{dg} is right Quillen. The right adjoint sends a dg-category to its dg-nerve, while the left adjoint freely generates a dg-category from a simplicial set.

In particular, a quasi-equivalence of dg-categories induces an equivalence of ∞ -categories, and a fibration induces a categorical fibration of dg-nerves.

The left adjoint is characterized by natural bijections

$$\text{Hom}_{\mathbf{Cat}_{\text{dg},k}}(L_{\text{dg}}(\kappa), C) \simeq \text{Hom}_{\mathbf{sSet}}(\kappa, N_*^{\text{dg}}(C))$$

for every simplicial set κ and dg-category C . Equivalently, it is the left Kan extension of its values on standard simplices:

$$L_{\text{dg}}(\kappa) \simeq \text{colim}_{\Delta^n \rightarrow \kappa} L_{\text{dg}}(\Delta^n)$$

Here $L_{\text{dg}}(\Delta^n)$ is the universal dg-category whose dg-functors to C are the n -simplices of $N_*^{\text{dg}}(C)$. This universal description is all that is needed for the Quillen adjunction; an explicit model can be obtained by applying normalized chains to simplicial rigidification.

Proof sketch. The dg-nerve preserves small limits and filtered colimits, so it admits a left adjoint. Suppose $F : C \rightarrow D$ is a fibration. In a lifting problem for an inner horn, the target filler determines the missing element in a mapping complex of D . Degreewise surjectivity lifts this element to C ; the same formula used in the inner-horn proof then supplies the remaining face. Thus $N_*^{\text{dg}}(F)$ is an inner fibration. The isofibration condition on hF lifts equivalences and makes it a categorical fibration.

If F is also a weak equivalence, the mapping-space formula above turns each quasi-isomorphism of mapping complexes into a weak homotopy equivalence, while the equivalence hF gives essential surjectivity. Hence $N_{\bullet}^{\text{dg}}(F)$ is a categorical equivalence. Therefore $N_{\bullet}^{\text{dg}}$ preserves fibrations and trivial fibrations and is right Quillen. $\square$

Remark (Summary of the Construction). The preceding discussion can be organized into the following chain of ideas.

1. A dg-category stores morphisms from x to y in the chain complex $\text{Map}_C(x, y)_*$, rather than in a set.
2. Connective truncation followed by Dold–Kan turns this complex into the mapping space of the dg-nerve:

$$\text{Map}_{N_{\bullet}^{\text{dg}}(C)}(x, y) \simeq \text{DK}(\tau_{\geq 0} \text{Map}_C(x, y)_*)$$

Consequently,

$$\pi_m \text{Map}_{N_{\bullet}^{\text{dg}}(C)}(x, y) \simeq H_m(\text{Map}_C(x, y)_*)$$

Thus the homology of a mapping complex becomes the homotopy of a mapping space. In degree zero this recovers $hN_{\bullet}^{\text{dg}}(C) \simeq hC$.

3. The direct dg-nerve and the homotopy coherent nerve of the associated simplicial category are equivalent. They are two models of the same ∞ -category: the former is explicit, while the latter explains the simplicial enrichment behind it.
4. The model structure on dg-categories is designed so that a quasi-equivalence is exactly a quasi-isomorphism on every mapping complex together with an equivalence on homotopy categories. The mapping-space formula therefore shows that the dg-nerve sends every quasi-equivalence to an equivalence of ∞ -categories.
5. These facts are compatible with the Quillen adjunction

$$L_{\text{dg}} : \mathbf{sSet}_{\text{Joyal}} \rightleftarrows \mathbf{Cat}_{\text{dg}, k} : N_{\bullet}^{\text{dg}}$$

Hence the dg-nerve is the bridge from differential graded algebra to ∞ -categories: chain homotopies and their higher compatibilities are retained as higher simplices. This statement gives only a Quillen *adjunction*. It does not assert that the two model categories are Quillen equivalent.

3.2 Derived ∞ -Categories

The dg-nerve remembers chain homotopies and all their higher coherences, but it does not yet identify quasi-isomorphic complexes. The derived ∞ -category is obtained by performing exactly this final localization.

3.2.1 Definition and Universal Property

Let A be an abelian category. Regard $\mathbf{Ch}(A)$ as the dg-category whose mapping complexes were defined above, and put

$$\mathbf{Ch}_\infty(A) := \mathbf{N}^{\mathrm{dg}}(\mathbf{Ch}(A))$$

Its homotopy category is the ordinary homotopy category $K(A)$ of chain complexes. Let W be the collection of quasi-isomorphisms, that is, chain maps $f : x \rightarrow y$ for which every map

$$H_n(f) : H_n(x) \rightarrow H_n(y)$$

is an isomorphism.

Definition 3.2.4 (Derived ∞ -Category). The *derived ∞ -category* of A is the ∞ -categorical localization

$$\mathbf{D}(A) := \mathbf{Ch}_\infty(A)[W^{-1}]$$

It has the same objects as $\mathbf{Ch}_\infty(A)$, but every quasi-isomorphism is made into an equivalence. We write

$$q : \mathbf{Ch}_\infty(A) \rightarrow \mathbf{D}(A)$$

for the localization functor.

The definition is characterized by the universal property

$$\mathbf{Fun}(\mathbf{D}(A), C) \simeq \mathbf{Fun}_W(\mathbf{Ch}_\infty(A), C)$$

where the right-hand side is the full subcategory of functors which carry every quasi-isomorphism to an equivalence. Thus a construction on complexes descends to $\mathbf{D}(A)$ precisely when it is invariant under quasi-isomorphisms; the descent and all its coherences are then unique up to a contractible space of choices.

Proposition 3.2.5 (Stable Verdier Quotient). The ∞ -category $\mathbf{Ch}_\infty(A)$ is stable. Let $\mathrm{Ac}(A)$ be its full stable subcategory of acyclic complexes. Then

$$\mathbf{D}(A) \simeq \mathbf{Ch}_\infty(A) / \mathrm{Ac}(A)$$

is a stable Verdier quotient, and q is exact. Moreover,

$$h\mathbf{D}(A) \simeq K(A) / \mathrm{Ac}(A)$$

is the classical triangulated derived category of A .

Proof sketch. Shifts and cofibers in $\mathbf{Ch}_\infty(A)$ are represented by shifts and mapping cones of complexes. A map f is a quasi-isomorphism exactly when $\mathrm{cofib}(f)$ is acyclic. Localizing at W is therefore the same as killing the stable subcategory $\mathrm{Ac}(A)$. A quotient of a stable ∞ -category by a stable subcategory is stable, and passage to the homotopy category gives the usual Verdier quotient. $\square$

Consequently, a cofiber sequence in $\mathbf{D}(A)$ is represented by a mapping-cone sequence

$$x \xrightarrow{f} y \rightarrow \mathrm{Cone}(f) \rightarrow x[1]$$

and becomes a distinguished triangle in $h\mathbf{D}(A)$. The ∞ -category retains the functorial cofiber and all higher coherences which its triangulated homotopy category forgets.

3.2.2 Homology, the t -Structure, and Ext

Proposition 3.2.6 (Standard t -Structure). The derived ∞ -category has a canonical t -structure defined by

$$\mathbf{D}(A)_{\geq 0} := \{x : H_n(x) = 0 \text{ for } n < 0\}$$

$$\mathbf{D}(A)_{\leq 0} := \{x : H_n(x) = 0 \text{ for } n > 0\}$$

Its truncation functors are induced by the good truncations of complexes, and its homology-object functors are the ordinary homology functors

$$\pi_n(x) \simeq H_n(x)$$

The inclusion of objects concentrated in degree zero induces an equivalence of abelian categories

$$A \simeq \mathbf{D}(A)^{\heartsuit}$$

Proof sketch. Good truncations give a functorial fiber-cofiber sequence

$$\tau_{\geq 0}x \rightarrow x \rightarrow \tau_{\leq -1}x$$

and the two terms have the required homology vanishing. If $x \in \mathbf{D}(A)_{\geq 0}$ and $y \in \mathbf{D}(A)_{\leq -1}$, every map $x \rightarrow y$ vanishes by truncation, giving orthogonality. A complex with homology only in degree zero is quasi-isomorphic to that homology object placed in degree zero, which identifies the heart with A . $\square$

The usual bounded variants are recovered directly from this t -structure:

$$\mathbf{D}^+(A) := \cup_{m \in \mathbb{Z}} \mathbf{D}(A)_{\geq m}, \quad \mathbf{D}^-(A) := \cup_{n \in \mathbb{Z}} \mathbf{D}(A)_{\leq n}$$

$$\mathbf{D}^b(A) := \cup_{m \leq n} (\mathbf{D}(A)_{\geq m} \cap \mathbf{D}(A)_{\leq n})$$

For $x, y \in \mathbf{D}(A)$ and $m \geq 0$, stability gives

$$\pi_m \mathrm{Map}_{\mathbf{D}(A)}(x, y) \simeq \mathrm{Hom}_h \mathbf{D}(A)(x[m], y)$$

For $a, b \in A \simeq \mathbf{D}(A)^{\heartsuit}$, define

$$\mathrm{Ext}_A^n(a, b) := \mathrm{Hom}_{h\mathbf{D}(A)}(a, b[n]) \quad (n \geq 0)$$

Equivalently,

$$\mathrm{Ext}_A^n(a, b) \simeq \pi_0 \mathrm{Map}_{\mathbf{D}(A)}(a, b[n])$$

This recovers the classical Yoneda Ext groups. Notice the direction of the shift: with our homological indexing, $\pi_m \mathrm{Map}_{\mathbf{D}(A)}(a, b) \simeq \mathrm{Hom}_{h\mathbf{D}(A)}(a, b[-m])$. Thus positive Ext is seen by shifting the target before taking the mapping space.

3.2.3 Resolutions and Derived Functors

The localization definition is conceptual, while resolutions make it computable.

Definition 3.2.5 (K -Projective and K -Injective Complexes). A complex p is K -projective if

$$\mathrm{Map}_{\mathbf{Ch}_{\infty}(A)}(p, a) \simeq *$$

for every acyclic complex a . Dually, a complex i is K -injective if

$$\mathrm{Map}_{\mathbf{Ch}_\infty(A)}(a, i) \simeq *$$

for every acyclic a .

Proposition 3.2.7 (Computing the Localization). If every complex x admits a quasi-isomorphism $p \rightarrow x$ with p K -projective, then the full dg-subcategory of K -projective complexes presents $\mathbf{D}(A)$. Dually, if every x admits a quasi-isomorphism $x \rightarrow i$ with i K -injective, then the K -injective complexes present $\mathbf{D}(A)$.

With either kind of replacement, derived mapping spaces can be computed by

$$\mathrm{Map}_{\mathbf{D}(A)}(x, y) \simeq \mathrm{DK}(\tau_{\geq 0} \underline{\mathrm{Hom}}_A(p, y))$$

$$\mathrm{Map}_{\mathbf{D}(A)}(x, y) \simeq \mathrm{DK}(\tau_{\geq 0} \underline{\mathrm{Hom}}_A(x, i))$$

Proof sketch. A quasi-isomorphism between K -projective complexes is a chain-homotopy equivalence, and the same holds for K -injectives. Moreover, mapping from a K -projective complex or into a K -injective complex already sends quasi-isomorphisms to equivalences. Hence the displayed mapping objects satisfy the universal property of the localization. $\square$

If A is a Grothendieck abelian category, every complex admits a K -injective resolution; in particular, $\mathbf{D}(A)$ is a presentable stable ∞ -category. For module categories one may also use K -projective resolutions.

Let $F : A \rightarrow B$ be additive and apply it degreewise to complexes. If F is exact, it preserves quasi-isomorphisms and therefore induces an exact functor

$$\mathbf{D}(F) : \mathbf{D}(A) \rightarrow \mathbf{D}(B)$$

For a nonexact functor, its total left and right derived functors, when they exist, are the Kan extensions through q . In terms of adapted resolutions, they are computed by

$$\mathrm{LF}(x) \simeq F(p), \quad \mathrm{RF}(x) \simeq F(i)$$

The point of a resolution is precisely that the right-hand side no longer depends on the chosen representative of x .

Example (Modules over a Ring). For a ring R , the derived ∞ -category is

$$\mathbf{D}(R) := \mathbf{D}(\mathbf{Mod}_R)$$

If $p \rightarrow m$ is a K -projective resolution and n is another complex, then

$$m \otimes_R^{\mathrm{L}} n \simeq p \otimes_R n, \quad \mathrm{RHom}_R(m, n) \simeq \underline{\mathrm{Hom}}_R(p, n)$$

When m, n are modules placed in degree zero, our homological convention gives

$$\mathrm{H}_r(m \otimes_R^{\mathrm{L}} n) \simeq \mathrm{Tor}_r^R(m, n), \quad \mathrm{H}_{-r}(\mathrm{RHom}_R(m, n)) \simeq \mathrm{Ext}_R^r(m, n)$$

Remark (Core Idea). The construction has two logically separate stages:

$$\mathbf{Ch}(A) \xrightarrow{\text{dg-nerve}} \mathbf{Ch}_\infty(A) \xrightarrow{\text{invert quasi-isomorphisms}} \mathbf{D}(A)$$

The first stage promotes chain homotopies to coherent higher morphisms. The second discards the choice of resolution by identifying complexes with the same derived information. Stability then organizes cones into cofiber sequences, the standard t -structure recovers homology and the original abelian category, and resolutions provide concrete formulas for mapping spaces and derived functors.

4 Spectra

4.1 Why We Need Spectra?

4.1.1 “Linear” Matters

Since the title of this book is actually “Linear Algebra”, we have to fix one thing: since the homotopy theory of pointed spaces is not “linear”. For pointed spaces X, Y , consider

$$[X, Y]_* := \pi_0 \text{Map}_*(X, Y)$$

to be the homotopy classes of pointed maps, it is only a set, not a group. Things changes if X is a suspension

$$X = \Sigma X'$$

then, through

$$\text{Map}_*(\Sigma X', Y) \simeq \text{Map}_*(X', \Omega Y)$$

We have

$$[\Sigma X', Y]_* = \pi_1 \text{Map}_*(X', Y)$$

naturally has a group structure. And if we suspension it again by

$$X = \Sigma^2 X''$$

then we have

$$[\Sigma^2 X'', Y]_* = \pi_2 \text{Map}_*(X'', Y)$$

which is an abelian group. In fact, by suspending the space enough times, making the homotopy theory more and more “linear”.

By this idea, we would like to consider

$$[X, Y]_* \rightarrow [\Sigma X, \Sigma Y]_* \rightarrow [\Sigma^2 X, \Sigma^2 Y]_* \rightarrow \dots$$

and take the limit

$$[X, Y]^s = \varinjlim_n [\Sigma^n X, \Sigma^n Y]$$

called the *stable homotopy classes of maps* from X to Y . For fine spaces, this forms an abelian group.

4.1.2 Inverting Suspension

There is a fatal problem exists in the suspension of pointed animae, which is that the suspension functor

$$\Sigma : \mathbf{Ani}_* \rightarrow \mathbf{Ani}_*$$

is not invertible, there does not exist a functor

$$\Sigma^{-1} : \mathbf{Ani}_* \rightarrow \mathbf{Ani}_*$$

corresponding to the inverse of suspension (i.e. we want the ∞ -category to be stable). Stable homotopy theory wants to invert the suspension functor, and this is the main reason why we need *spectra*, which is a world where suspension forms an equivalence.

$$\Sigma : \mathbf{Sp} \xrightarrow{\sim} \mathbf{Sp}$$

just like the $[1]$ functor on the derived category of an abelian category.

4.2 Brown Representability Theorem

From now on, we will build a relation between cohomology theory and the theory of spectra, we will start with the Brown representability theorem, which is a very important theorem in stable homotopy theory.

4.2.1 Generalized Cohomology Theories

Definition 4.2.1 (Reduced Generalized Cohomology Theory). A *reduced generalized cohomology theory* on pointed CW complexes is a family of contravariant functors

$$E^q : (h\mathbf{CW}_*)^{\text{op}} \rightarrow \mathbf{Ab} \quad (q \in \mathbb{Z})$$

with the following properties.

1. *Reducedness.* $E^q(*) = 0$ for every q .
2. *Suspension.* There are natural isomorphisms

$$E^q(x) \simeq E^{q+1}(\Sigma x)$$

3. *Exactness.* Every cofiber sequence

$$a \rightarrow x \rightarrow x/a \rightarrow \Sigma a$$

induces a natural long exact sequence

$$\dots \rightarrow E^q(x/a) \rightarrow E^q(x) \rightarrow E^q(a) \rightarrow E^{q+1}(x/a) \rightarrow \dots$$

4. *Wedge axiom.* For every family $\{x_\alpha\}$,

$$E^q\left(\bigvee_\alpha x_\alpha\right) \simeq \prod_\alpha E^q(x_\alpha)$$

The adjective *generalized* means that no dimension axiom is imposed.

Thus E^q is the degree- q cohomology functor, not the q th power of a single functor. Exactness implies the Mayer–Vietoris gluing property, while the wedge axiom is already one of the Brown axioms. Consequently, after forgetting the abelian-group structure, every E^q is a Brown functor.

Example. Reduced singular cohomology $\tilde{H}^q(-; G)$ is the basic example. Its dimension axiom says

$$\tilde{H}^0(S^0; G) \simeq G, \quad \tilde{H}^q(S^0; G) \simeq 0 \quad (q \neq 0)$$

Generalized theories such as topological K -theory need not satisfy this axiom.

Remark (Homology versus Cohomology). A reduced generalized *homology* theory is instead a covariant family

$$E_q : h\mathbf{CW}_* \rightarrow \mathbf{Ab}$$

which sends wedges to direct sums and cofiber sequences to long exact sequences. Brown representability below directly concerns the contravariant cohomology functors E^q . Homology theories acquire their representing description after passing to spectra.

4.2.2 Classical Version

Let $h\mathbf{CW}_*$ denote the homotopy category of pointed CW complexes.

Definition 4.2.2 (Brown Functor). A contravariant functor

$$F : (h\mathbf{CW}_*)^{\text{op}} \rightarrow \mathbf{Set}_*$$

is a *Brown functor* if it satisfies:

1. *Wedge axiom.*

$$F\left(\bigvee_{\alpha} x_{\alpha}\right) \simeq \prod_{\alpha} F(x_{\alpha})$$

2. *Mayer–Vietoris axiom.* If $x = a \cup b$ is a union of pointed subcomplexes whose interiors cover x , then

$$F(x) \rightarrow F(a) \times_{F(a \cap b)} F(b)$$

is surjective.

Every representable functor $[-, y]_*$ has these properties. The wedge axiom is the coproduct universal property, while Mayer–Vietoris follows from the homotopy extension property for inclusions of CW subcomplexes.

Theorem 4.2.1 (Classical Brown Representability). Every Brown functor is representable. More precisely, there exist a pointed CW complex y and a universal class $u \in F(y)$ such that

$$T_u : [x, y]_* \rightarrow F(x), \quad [f] \mapsto f^*u$$

is a natural bijection for every pointed CW complex x . The representing space y is unique up to pointed homotopy equivalence.

Proof sketch. The representing pair (y, u) is built by cells. First wedge onto y enough spheres so that every class in $F(S^n)$ is the pullback of u ; this forces surjectivity on spheres. Next attach cells along every map $S^n \rightarrow y$ which pulls u back to the basepoint; the Mayer–Vietoris axiom extends u across these cells and kills the kernel. Repeating this in every dimension and taking a mapping telescope gives

$$[S^n, y]_* \simeq F(S^n)$$

for all n .

A relative version of the same cell construction extends a map $a \rightarrow y$ from a CW subcomplex $a \subset x$ whenever its pullback of u agrees with a prescribed class in $F(x)$. Taking $a = *$ proves that T_u is surjective. Applying the relative statement to the two ends of a cylinder shows that two maps with the same pullback of u are homotopic, so T_u is injective. Whitehead's theorem upgrades the weak equivalences arising in the construction to homotopy equivalences, and Yoneda gives uniqueness of y . $\square$

Remark (From Cohomology Theories to Spectra). Let $\{E^q\}_{q \in \mathbb{Z}}$ be a reduced generalized cohomology theory on pointed CW complexes. Each functor E^q satisfies the wedge and Mayer–Vietoris axioms, so Brown representability gives pointed CW complexes e_q with

$$E^q(x) \simeq [x, e_q]_*$$

The suspension isomorphism and the suspension–loop adjunction give

$$[x, e_q]_* \simeq [\Sigma x, e_{q+1}]_* \simeq [x, \Omega e_{q+1}]_*$$

Yoneda therefore yields equivalences

$$e_q \simeq \Omega e_{q+1}$$

Hence the representing spaces assemble into an Ω -spectrum. This is the classical reason that spectra represent generalized cohomology theories.

4.2.3 Representability in an ∞ -Category

Let C be an ∞ -category. A space-valued presheaf

$$P : C^{\text{op}} \rightarrow \mathbf{Ani}$$

is *representable* if there is an object $y \in C$ and a natural equivalence

$$P(x) \simeq \text{Map}_C(x, y)$$

for every $x \in C$. By the ∞ -categorical Yoneda lemma, choosing such an equivalence is the same as choosing a *universal element* $u \in P(y)$ for which evaluation induces the displayed equivalences. The representing object is unique up to equivalence.

Passing to connected components gives the ordinary representable functor

$$\text{Hom}_{hC}(-, y) \simeq \pi_0 \text{Map}_C(-, y)$$

on the homotopy category. Accordingly, a functor $F : (hC)^{\text{op}} \rightarrow \mathbf{Set}$ is representable when there are $y \in C$ and $u \in F(y)$ such that

$$\text{Hom}_{hC}(x, y) \rightarrow F(x), \quad [f] \mapsto f^*u$$

is a bijection for every x . The theorem below concerns this $\mathbf{Set}$ -valued notion: it reconstructs the representing object from the behavior of F on colimits in the underlying ∞ -category.

For an ordinary presentable category D , the adjoint functor theorem says that $F : D^{\text{op}} \rightarrow \mathbf{Set}$ is representable exactly when it carries colimits in D to limits of sets. The difficulty here is that hC

need not possess the colimits which exist in C . Brown representability replaces that unavailable criterion by conditions imposed directly on coproducts and pushouts in C .

Remark (Basic Principle). A representable contravariant functor turns colimits into limits, since

$$\mathrm{Map}_C\left(\mathrm{colim}_i x_i, y\right) \simeq \lim_i \mathrm{Map}_C(x_i, y)$$

After applying π_0 , arbitrary coproducts still become products. For a pushout, connected components retain the existence, but not necessarily the uniqueness, of a gluing; this explains the surjectivity condition in Brown representability.

4.2.4 Cogroup Objects and Compact Generators

Definition 4.2.3 (Cogroup Object). Let D be a category with finite coproducts. A *cogroup object* is an object $s \in D$ equipped with a comultiplication

$$\Delta : s \rightarrow s \sqcup s$$

together with a counit and coinverse, such that for every $y \in D$ the induced operation

$$\mathrm{Hom}_D(s, y) \times \mathrm{Hom}_D(s, y) \simeq \mathrm{Hom}_D(s \sqcup s, y) \xrightarrow{\Delta^*} \mathrm{Hom}_D(s, y)$$

is a group law. Equivalently, $\mathrm{Hom}_D(s, -)$ naturally takes values in groups.

The guiding example is a suspension. If C admits finite colimits and $x \rightarrow \emptyset$ is a map to its initial object, then

$$\Sigma x := \emptyset \sqcup_x \emptyset$$

is a cogroup object of hC . Indeed, $\mathrm{Hom}_{hC}(\Sigma x, y)$ is a fundamental group of a mapping space, and its group operation is loop concatenation. In a stable ∞ -category, hC is additive, so every object is a cogroup object; the comultiplication is the diagonal $s \rightarrow s + s$ and the coinverse is $-\mathrm{id}_s$.

Definition 4.2.4 (Compact Generators). An object $s \in C$ is *compact* if $\mathrm{Map}_C(s, -)$ preserves filtered colimits. A set $\{s_\alpha\}_{\alpha \in A}$ *generates C under small colimits* if the smallest full subcategory containing every s_α and closed under equivalences and small colimits is all of C .

Compactness lets maps out of s_α detect a colimit at a finite stage; generation means that the functors $\mathrm{Map}_C(s_\alpha, -)$ jointly detect equivalences. These properties abstract the roles played by spheres and CW cells in the classical proof.

4.2.5 General Brown Representability

Theorem 4.2.2 (Brown Representability). Let C be a presentable ∞ -category containing a set $\{s_\alpha\}_{\alpha \in A}$ such that:

1. every s_α is a cogroup object of hC ;
2. every s_α is compact;
3. the objects s_α generate C under small colimits.

Then a functor $F : (hC)^{\text{op}} \rightarrow \mathbf{Set}$ is representable if and only if it satisfies the following conditions.

1. For every collection $\{c_\beta\}$, the canonical map

$$F(\sqcup_\beta c_\beta) \rightarrow \prod_\beta F(c_\beta)$$

is a bijection.

2. For every pushout $d' \simeq c' \sqcup_c d$, the canonical map

$$F(d') \rightarrow F(c') \times_{F(c)} F(d)$$

is surjective.

Proof sketch. Necessity follows from the mapping-space formula above. For sufficiency, begin with a pair (x, u) and coproduct copies of the generators until

$$\text{Hom}_{hC}(s_\alpha, x) \rightarrow F(s_\alpha)$$

is surjective for every α . Use pushouts to kill its kernels, and iterate. Cogroup structures make these kernels groups, compactness lets maps from s_α factor through a finite stage, and generation shows that the resulting universal pair represents F on every object. This is exactly the classical argument *add generators, then kill relations*, with compact cogroup generators replacing spheres.

□

Corollary 4.2.1. Every compactly generated presentable stable ∞ -category satisfies the hypotheses of Brown representability, because every object of its homotopy category is a cogroup object. In particular, a contravariant functor on its homotopy category is representable exactly when it takes coproducts to products and pushouts to weak pullbacks of sets.

For connected pointed spaces, the compact cogroup generator is S^1 ; its suspensions detect all higher homotopy groups. Thus the generalized theorem recovers the classical Brown theorem, while its stable case is the form that will be used for spectra.

Remark (From Cohomology to Homotopy). Brown representability turns a reduced generalized cohomology theory into homotopy-theoretic data. If e_q represents E^q , then

$$E^q(x) \simeq [x, e_q]_*, \quad E^q(S^m) \simeq \pi_m(e_q)$$

Thus cohomology classes are homotopy classes of maps, and the coefficient groups of the theory are homotopy groups of the representing spaces. Moreover, Yoneda identifies a natural cohomology operation $E^q \rightarrow E''$ with a homotopy class of maps $e_q \rightarrow e'_r$.

The suspension axiom organizes the spaces e_q into an Ω -spectrum, so a whole cohomology theory becomes a single stable homotopy type. Brown's theorem directly concerns contravariant cohomology theories. Covariant homology theories enter after stabilization: a representing spectrum e defines homology by the homotopy groups of $e \wedge \Sigma^\infty x$.

4.3 Spectrum Objects

Spectrum objects are the basic device for extracting stable information from an unstable ∞ -category. The key input is *excision*: a linear functor should turn a gluing problem into a pullback problem.

4.3.1 Reduced Excisive Functors

The classical model is singular homology. The simplicial abelian group $\mathbb{Z}\text{Sing}(x)_\bullet$ is a Kan complex and

$$H_n(x; \mathbb{Z}) \simeq \pi_n(\mathbb{Z}\text{Sing}(x)_\bullet) \quad (n \geq 0)$$

For an open cover $x = u \cup v$, the Mayer–Vietoris sequence is obtained by applying homotopy groups to the homotopy pullback square associated with u , v , $u \cap v$, and x . This is the prototype for the following definition.

Definition 4.3.5 (Reduced Excisive Functor). Let C admit pushouts and let D admit pullbacks. A functor $F : C \rightarrow D$ is *excisive* if it carries every pushout square in C to a pullback square in D .

If C and D have final objects, then F is *reduced* if it carries a final object of C to a final object of D . We write

$$\mathbf{Exc}_*(C, D) \subseteq \mathbf{Fun}(C, D)$$

for the full subcategory of reduced excisive functors.

Remark (Excision Is Exactness). If C and D are stable, a functor $F : C \rightarrow D$ is reduced and excisive if and only if it is exact. Thus excision is the unstable formulation of the familiar requirement that a linear functor preserve exact triangles.

Equivalently, F is exact precisely when it preserves zero objects and the canonical map

$$\Sigma_D F(x) \rightarrow F(\Sigma_C x)$$

is an equivalence for every x .

There is a useful one-object test for excision. Let C be pointed with finite colimits, let D be pointed with finite limits, and let $F : C \rightarrow D$ be reduced. Applying F to the suspension square of x produces a canonical map

$$\eta_x : F(x) \rightarrow \Omega_D F(\Sigma_C x)$$

Proposition 4.3.1 (Suspension Test for Excision). The functor F is excisive if and only if η_x is an equivalence for every $x \in C$.

Proof sketch. If F is excisive, it sends the suspension pushout of x to the pullback defining $\Omega_D F(\Sigma_C x)$. Conversely, compare an arbitrary pushout square with the four suspension squares of its vertices. The maps η_x identify the resulting comparison map with an equivalence, so the image square is a pullback. $\square$

4.3.2 The Definition of a Spectrum Object

Let $\mathbf{Ani}_*^{\text{fin}}$ be the ∞ -category of finite pointed anima. It is generated by S^0 under finite colimits.

Definition 4.3.6 (Spectrum Object). Let C be an ∞ -category with finite limits. A *spectrum object* of C is a reduced excisive functor

$$x : \mathbf{Ani}_*^{\text{fin}} \rightarrow C$$

The full ∞ -category of spectrum objects is

$$\mathbf{Sp}(C) := \mathbf{Exc}_*(\mathbf{Ani}_*^{\text{fin}}, C)$$

Remark (Why Does This Definition Produce Spectra?). At first sight, a spectrum object is merely a functor in $\mathbf{Fun}(\mathbf{Ani}_*^{\text{fin}}, C)$. The reduced and excisive conditions force this functor to contain an infinite sequence of deloopings.

For every $n \geq 0$, the suspension square

$$\begin{array}{ccc} S^n & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & \Sigma S^n \simeq S^{n+1} \end{array}$$

is a pushout in $\mathbf{Ani}_*^{\text{fin}}$. Applying x gives

$$\begin{array}{ccc} x(S^n) & \longrightarrow & x(*) \\ \downarrow & & \downarrow \\ x(*) & \longrightarrow & x(S^{n+1}) \end{array}$$

Reducedness says $x(*)$ is terminal, while excision says that the second square is a pullback. Therefore

$$x(S^n) \simeq * \times_{x(S^{n+1})} * \simeq \Omega x(S^{n+1})$$

Setting $x_n := x(S^n)$ gives coherent equivalences

$$x_n \simeq \Omega x_{n+1}$$

Hence one reduced excisive functor automatically contains the infinite delooping data

$$x_0 \simeq \Omega x_1, \quad x_1 \simeq \Omega x_2, \quad x_2 \simeq \Omega x_3, \quad \dots$$

The functor

$$\Omega^\infty : \mathbf{Sp}(C) \rightarrow C, \quad x \mapsto x_0$$

is evaluation at S^0 . More precisely, if C_* denotes the ∞ -category of pointed objects of C , then

$$\mathbf{Sp}(C) \simeq \lim \left(\dots \xrightarrow{\Omega} C_* \xrightarrow{\Omega} C_* \right)$$

When C is already pointed, this becomes the tower displayed at the beginning of this section.

Remark (Finite versus Compact). The category $\mathbf{Ani}_*^{\text{fin}}$ need not contain every compact pointed anima. The full subcategory of compact objects of $\mathbf{Ani}_*$ is its idempotent completion. The finite category is the one used above because of its universal property under finite colimits.

4.3.3 Classical Sequential Spectra

The classical point-set model makes the tower of deloopings explicit.

Definition 4.3.7 (Sequential Spectrum). A *sequential prespectrum* is a sequence of pointed spaces $(x_0, x_1, \dots)$ equipped with structure maps

$$\sigma_n : \Sigma x_n \rightarrow x_{n+1}$$

Equivalently, by adjunction, it has maps

$$\tilde{\sigma}_n : x_n \rightarrow \Omega x_{n+1}$$

It is an Ω -spectrum if every $\tilde{\sigma}_n$ is a weak homotopy equivalence.

For $k \in \mathbb{Z}$, the stable homotopy group of a prespectrum x is

$$\pi_k^s(x) := \varinjlim_n \pi_{k+n}(x_n)$$

where n is taken sufficiently large that $k + n \geq 0$. A map is a *stable equivalence* if it induces an isomorphism on every π_k^s . The classical stable homotopy category is obtained by inverting these maps. Every sufficiently well-behaved prespectrum admits a spectrification $x \rightarrow qx$ to an Ω -spectrum with the same stable homotopy groups.

Example (Basic Spectra). For a pointed space x , its suspension prespectrum has levels

$$(\Sigma^\infty x)_n := \Sigma^n x$$

After spectrification this defines the suspension spectrum $\Sigma^\infty x \in \mathbf{Sp}$. In particular,

$$\mathbb{S} := \Sigma^\infty S^0$$

is the sphere spectrum, and its homotopy groups are the stable stems

$$\pi_k(\mathbb{S}) \simeq \varinjlim_n \pi_{k+n}(S^n)$$

If a is an abelian group, the Eilenberg–Mac Lane spectrum Ha has spaces $K(a, n)$ and structure equivalences

$$K(a, n) \simeq \Omega K(a, n+1)$$

It is characterized by $\pi_0(Ha) \simeq a$ and $\pi_k(Ha) \simeq 0$ for $k \neq 0$. Ordinary reduced cohomology is then represented by

$$\tilde{H}^q(x; a) \simeq [\Sigma^\infty x, \Sigma^q Ha]_{\mathbf{Sp}}$$

Remark (Classical versus ∞ -Categorical Spectra). The two descriptions encode the same stable homotopy theory.

classical model	∞ -categorical model
spaces x_n and maps $\Sigma x_n \rightarrow x_{n+1}$	a functor $\mathbf{Ani}_*^{\text{fin}} \rightarrow \mathbf{Ani}$
Ω -spectrum condition	reduced excision
invert stable equivalences	equivalences already encode localization
chosen point-set structure maps	all homotopy coherences retained

More precisely, localizing any standard model category of classical spectra at its stable equivalences presents the ∞ -category $\mathbf{Sp} \simeq \mathbf{Sp}(\mathbf{Ani})$. Thus an ∞ -categorical spectrum should be compared with the stable homotopy type of a classical spectrum, not with one chosen point-set presentation.

4.3.4 Stability

Theorem 4.3.3 (Stability Criterion). For a pointed ∞ -category C , the following conditions are equivalent.

1. The ∞ -category C is stable.
2. It admits finite colimits and $\Sigma_C : C \rightarrow C$ is an equivalence.
3. It admits finite limits and $\Omega_C : C \rightarrow C$ is an equivalence.

Proof sketch. In a stable ∞ -category, pushouts and pullbacks agree, so suspension and looping are inverse equivalences. Conversely, if looping is invertible, every pullback can be delooped into a pushout; hence pullbacks and pushouts agree. The suspension version is dual. $\square$

Theorem 4.3.4 (Stability of Excisive Functors). Let C be pointed and admit finite colimits, and let D admit finite limits. Then $\mathbf{Exc}_*(C, D)$ is a stable ∞ -category. In particular, $\mathbf{Sp}(D)$ is stable.

Proof sketch. Reduced excisive functors are closed under finite limits, which are computed pointwise. Their loop functor is also pointwise, and the suspension test identifies it with an equivalence. The stability criterion therefore applies. Taking $C = \mathbf{Ani}_*^{\text{fin}}$ gives the assertion for spectrum objects. $\square$

Corollary 4.3.2 (Recognizing Stability by Spectrum Objects). Let C admit finite limits. Then C is stable if and only if evaluation at S^0 is an equivalence

$$\Omega_C^\infty : \mathbf{Sp}(C) \xrightarrow{\sim} C$$

In this case, we denote its inverse by

$$\Sigma_C^\infty : C \xrightarrow{\sim} \mathbf{Sp}(C)$$

Thus

$$\Omega_C^\infty \Sigma_C^\infty \simeq \text{id}_C, \quad \Sigma_C^\infty \Omega_C^\infty \simeq \text{id}_{\mathbf{Sp}(C)}$$

For $c \in C$, the spectrum $\Sigma_C^\infty c$ is determined levelwise by

$$(\Sigma_C^\infty c)_n \simeq \Sigma_C^n c$$

Proof sketch. If C is stable, then Ω_C is an equivalence, so the tower

$$\dots \xrightarrow{\Omega_C} C \xrightarrow{\Omega_C} C$$

is essentially constant. Its limit is therefore C , and evaluation at the zeroth level is an equivalence. Conversely, $\mathbf{Sp}(C)$ is stable; hence an equivalence $\mathbf{Sp}(C) \simeq C$ transports the stable structure to C . $\square$

Remark. In general, Σ^∞ denotes the suspension-spectrum or stabilization functor and is left adjoint to Ω^∞ when this adjoint exists. It is only when C is already stable that this adjunction becomes an adjoint equivalence and Σ_C^∞ is literally the inverse of Ω_C^∞ .

4.3.5 The Universal Property of Stabilization

Theorem 4.3.5 (Universal Property). Let C be pointed and admit finite colimits, and let D admit finite limits. Composition with evaluation at S^0 induces an equivalence

$$\mathbf{Exc}_*(C, \mathbf{Sp}(D)) \xrightarrow{\sim} \mathbf{Exc}_*(C, D)$$

Hence every reduced excisive functor $C \rightarrow D$ has an essentially unique refinement whose values are coherent spectrum objects. Taking $C = \mathbf{Ani}_*^{\text{fin}}$ shows in particular that stabilization is idempotent

$$\mathbf{Sp}(\mathbf{Sp}(D)) \simeq \mathbf{Sp}(D)$$

Remark (Formal Properties). Spectrum objects and their limits are computed pointwise. Consequently, for every simplicial set K there is a natural equivalence

$$\mathbf{Sp}(\mathbf{Fun}(K, C)) \simeq \mathbf{Fun}(K, \mathbf{Sp}(C))$$

If C is presentable, the reduced and excisive conditions define an accessible localization of a functor category. Hence $\mathbf{Sp}(C)$ is again presentable, as well as stable.

Remark (Homology Theories). A reduced spectrum-valued homology theory is, at its core, a reduced excisive functor

$$E : \mathbf{Ani}_*^{\text{fin}} \rightarrow \mathbf{Sp}$$

and its graded groups are recovered by

$$E_n(x) := \pi_n E(x)$$

Excision produces Mayer–Vietoris long exact sequences, while preservation of arbitrary wedges supplies the usual wedge axiom when the theory is extended from finite pointed

animae to all pointed animae. Thus the spectrum packages all degrees and all connecting maps of a homology theory into one object.

4.4 ∞ -Category of Spectra

4.4.1 Spectrum

The most important example of a stable ∞ -category is the category of spectra itself. The preceding construction turns the classical sequence of spaces and deloopings into the following intrinsic definition.

Definition 4.4.8 (The ∞ -Category of Spectra). A *spectrum* is a spectrum object of the ∞ -category of animae. We write

$$\mathbf{Sp} := \mathbf{Sp}(\mathbf{Ani}) \simeq \mathbf{Sp}(\mathbf{Ani}_*)$$

for the ∞ -category of spectra. Equivalently,

$$\mathbf{Sp} \simeq \mathbf{Exc}_*(\mathbf{Ani}_*^{\text{fin}}, \mathbf{Ani})$$

Proposition 4.4.2 (Equivalent Sequential Description). Equivalently, a spectrum is a sequence of pointed animae

$$x = (x_0, x_1, x_2, \dots)$$

together with coherent equivalences

$$\varepsilon_n : x_n \xrightarrow{\sim} \Omega x_{n+1} \quad (n \geq 0)$$

A morphism $f : x \rightarrow y$ is a sequence of pointed maps $f_n : x_n \rightarrow y_n$ compatible, up to coherent homotopy, with the equivalences ε_n . In particular,

$$\mathbf{Sp} \simeq \lim \left(\dots \xrightarrow{\Omega} \mathbf{Ani}_* \xrightarrow{\Omega} \mathbf{Ani}_* \right)$$

and the infinite-loop-space functor is evaluation at the zeroth term

$$\Omega^\infty x \simeq x_0$$

Remark. This is the intrinsic ∞ -categorical version of a classical Ω -spectrum. The word *coherent* is essential: besides the equivalences $x_n \simeq \Omega x_{n+1}$, the ∞ -categorical limit keeps all higher compatibilities automatically. Thus one may safely remember a spectrum as the infinite delooping sequence

$$x_0 \simeq \Omega x_1, \quad x_1 \simeq \Omega x_2, \quad x_2 \simeq \Omega x_3, \quad \dots$$

Definition 4.4.9 (Sphere Spectrum). The *sphere spectrum* is the suspension spectrum of the pointed 0-sphere

$$\mathbb{S} := \Sigma^\infty S^0 \in \mathbf{Sp}$$

In the classical sequential model it is represented by the suspension prespectrum

$$S^0, S^1, S^2, \dots$$

with the canonical structure maps $\Sigma S^n \simeq S^{n+1}$, followed by spectrification. Its homotopy groups are the stable stems

$$\pi_k(\mathbb{S}) \simeq \varinjlim_n \pi_{k+n}(S^n)$$

Remark (The Stable Homotopy Analogue of $\mathbb{Z}$). The sphere spectrum plays the role of the integers in stable homotopy theory. The analogy is structural:

- $\mathbb{Z}$ is the tensor unit in $\mathbf{Ab}$, while $\mathbb{S}$ is the smash-product unit in $\mathbf{Sp}$

$$\mathbb{Z} \otimes a \simeq a, \quad \mathbb{S} \wedge x \simeq x$$

- $\mathbb{Z}$ is the free rank-one generator of $\mathbf{Ab}$, while $\mathbb{S}$ is a compact generator of $\mathbf{Sp}$; maps from its shifts recover every homotopy group

$$\mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^n \mathbb{S}, x) \simeq \pi_n(x)$$

- Their endomorphisms in degree zero agree

$$\mathrm{End}_{h\mathbf{Sp}}(\mathbb{S}) \simeq \pi_0(\mathbb{S}) \simeq \mathbb{Z}$$

However, $\mathbb{S}$ is not the Eilenberg–Mac Lane spectrum HZ . The latter has only one nonzero homotopy group, whereas the higher stable stems $\pi_n(\mathbb{S})$ retain genuinely homotopical information. Thus $\mathbb{S}$ is the homotopically enriched analogue of $\mathbb{Z}$, not merely $\mathbb{Z}$ regarded as a spectrum.

Remark (Relation with the Classical Category). The homotopy category $h\mathbf{Sp}$ is the classical stable homotopy category. In particular, its morphisms are stable homotopy classes of maps. The various point-set models of spectra give different presentations, but after inverting stable equivalences they all present the same ∞ -category $\mathbf{Sp}$.

Remark (A Spectrum as a Homology Theory). Let $e \in \mathbf{Sp}$. Under the definition above, e is a reduced excisive functor

$$e : \mathbf{Ani}_*^{\mathrm{fin}} \rightarrow \mathbf{Ani}$$

For a map $a \rightarrow x$ of finite animae, form the pointed quotient

$$x/a := x \sqcup_a *$$

The relative groups associated with e are

$$e_n(x, a) := \pi_n(e(x/a)) \quad (n \geq 0)$$

where the base point of $e(x/a)$ is induced by $e(*) \rightarrow e(x/a)$. Negative degrees are recovered by shifting the spectrum e .

Reducedness gives $e_n(*) = 0$. Excision turns homotopy pushout squares of finite pointed anima into homotopy pullback squares, so applying homotopy groups produces excision, Mayer–Vietoris sequences, and the connecting maps of a generalized homology theory. Thus a spectrum is not merely a sequence of spaces: it coherently packages all degrees of a homology theory.

Corollary 4.4.3. The ∞ -category $\mathbf{Sp}$ is stable. Its suspension and loop functors are inverse equivalences

$$\Sigma : \mathbf{Sp} \xrightarrow{\sim} \mathbf{Sp} : \Omega$$

The stable ∞ -category $\mathbf{Sp}$ also carries a canonical Postnikov t -structure, organized by the vanishing of its homotopy groups. This will relate spectra to ordinary abelian groups through the heart of $\mathbf{Sp}$.

4.4.2 The Postnikov t -Structure

For $x \in \mathbf{Sp}$ and $n \in \mathbb{Z}$, define its n th homotopy group by

$$\pi_n(x) := \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^n \mathbb{S}, x)$$

The infinite loop-space structure makes this an abelian group in every degree, including degrees 0 and 1.

Proposition 4.4.3 (Postnikov t -Structure on Spectra). The full subcategories

$$\mathbf{Sp}_{\geq 0} := \{x \in \mathbf{Sp} : \pi_n(x) = 0 \text{ for } n < 0\}$$

and

$$\mathbf{Sp}_{\leq 0} := \{x \in \mathbf{Sp} : \pi_n(x) = 0 \text{ for } n > 0\}$$

determine an accessible t -structure on $\mathbf{Sp}$. It is both left and right complete.

If a spectrum x is represented by compatible spaces $x_n \simeq \Omega x_{n+1}$, then

$$x \in \mathbf{Sp}_{\leq m} \Leftrightarrow x_n \text{ is } (n+m)\text{-truncated for every } n \geq 0$$

Thus the truncations of a spectrum are ordinary Postnikov truncations, performed compatibly at every level.

Corollary 4.4.4 (The Heart of Spectra). The heart of the Postnikov t -structure is canonically equivalent to the category of abelian groups

$$\mathbf{Sp}^\heartsuit \simeq \mathbf{Ab}$$

Under this equivalence, an abelian group a corresponds to the Eilenberg–Mac Lane spectrum Ha . Its n th space is $K(a, n)$, and its only nonzero homotopy group is

$$\pi_0(Ha) \simeq a$$

Proof sketch. A spectrum in the heart has each level x_n both $(n - 1)$ -connected and n -truncated, hence $x_n \simeq K(a, n)$ for one abelian group a . Conversely, the spaces $K(a, n)$ form an Ω -spectrum. Completeness follows because a spectrum is reconstructed from its compatible Postnikov truncations; a spectrum whose homotopy groups all vanish is therefore zero. $\square$

Remark (The General Construction). More generally, if C is presentable, then $\mathbf{Sp}(C)$ has an accessible t -structure whose negative part is characterized by

$$x \in \mathbf{Sp}(C)_{\leq -1} \Leftrightarrow \Omega^\infty x \text{ is final in } C$$

Equivalently, its connective part is generated under colimits and extensions by the suspension spectra of objects of C .

4.4.3 Homotopy Groups and Compact Generation

Corollary 4.4.5 (Stable Whitehead Theorem). A map $f : x \rightarrow y$ of spectra is an equivalence if and only if

$$\pi_n(f) : \pi_n(x) \rightarrow \pi_n(y)$$

is an isomorphism for every $n \in \mathbb{Z}$.

Proof sketch. Apply the long exact sequence of homotopy groups to $\text{fib}(f)$. If every $\pi_n(f)$ is an isomorphism, then all homotopy groups of $\text{fib}(f)$ vanish. Completeness of the Postnikov t -structure forces $\text{fib}(f) \simeq 0$. $\square$

Proposition 4.4.4 (Compact Generation). The sphere spectrum $\mathbb{S}$ is a compact generator of $\mathbf{Sp}$ in the stable sense: its shifts $\{\Sigma^n \mathbb{S}\}_{n \in \mathbb{Z}}$ jointly detect equivalences. Consequently, $\mathbf{Sp}$ is compactly generated, and its compact objects are precisely the retracts of finite cell spectra.

Proof sketch. Maps from the shifts of $\mathbb{S}$ compute the homotopy groups

$$\text{Hom}_{h\mathbf{Sp}}(\Sigma^n \mathbb{S}, x) \simeq \pi_n(x)$$

so generation follows from the stable Whitehead theorem. Compactness is the fact that homotopy groups commute with filtered colimits. The compact objects are then obtained from $\mathbb{S}$ by finite cofibers, shifts, and retracts. $\square$

Remark. The proposition above is basically telling that the entire $\mathbf{Sp}$ is generated by $\mathbb{S}$ via suspensions, colimits and retracts.

Remark (Colimits and Infinite Loop Spaces). The functors $\pi_n : \mathbf{Sp} \rightarrow \mathbf{Ab}$ preserve products and filtered colimits. The infinite-loop-space functor has the stronger connective statement

$$\Omega^\infty : \mathbf{Sp}_{\geq 0} \rightarrow \mathbf{Ani}$$

preserves sifted colimits. Outside the connective part, Ω^∞ is a right adjoint and should not be expected to preserve arbitrary colimits.

4.5 Presentable Stable ∞ -Categories

Presentability is especially simple in the stable setting: finite colimits are already built into stability, so arbitrary coproducts control all small colimits.

4.5.1 Coproducts Control Colimits

Proposition 4.5.5 (Colimits in a Stable ∞ -Category). Let C be stable.

1. The category C admits all small colimits if and only if it admits all small coproducts.
2. If C and D admit small colimits, an exact functor $F : C \rightarrow D$ preserves all small colimits if and only if it preserves small coproducts.
3. If these conditions hold, an object $x \in C$ is compact if and only if every map

$$x \rightarrow \sqcup_{\alpha \in A} y_\alpha$$

factors, up to homotopy, through a finite subcoproduct.

Proof sketch. In a stable category, finite colimits are finite limits and are therefore already available. General colimits can be assembled from coproducts and cofibers, proving the first two statements. For the last, write a coproduct as the filtered colimit of its finite subcoproducts and apply the definition of compactness. $\square$

Thus, for exact functors, the apparently global problem of preserving every colimit reduces to checking only coproducts.

4.5.2 Presentability and Generators

Recall that an object $g \in C$ *generates* a stable ∞ -category if

$$\pi_0 \text{Map}_C(g, x) = * \Rightarrow x \simeq 0$$

Equivalently, all shifts of g jointly detect equivalences. A κ -compact object is one whose mapping-space functor preserves κ -filtered colimits.

Theorem 4.5.6 (Presentability Criterion). A stable ∞ -category C is presentable if and only if:

1. C admits small coproducts;
2. its homotopy category hC is locally small;
3. for some regular cardinal κ , it has a κ -compact generator g .

Proof sketch. A presentable category has a small family of compact objects; their coproduct gives one κ -compact generator for sufficiently large κ . Conversely, close the shifts of g under κ -small

colimits. This gives a small stable subcategory whose κ -Ind-completion maps fully faithfully to C . Since g detects zero objects, this map is an equivalence. $\square$

Remark. This criterion is visible entirely in hC . It is therefore the stable ∞ -categorical counterpart of compact-generation criteria for triangulated categories.

4.5.3 Stabilization and Its Presentable Universal Property

Theorem 4.5.7 (Presentability of Stabilization). If C is presentable, then $\mathbf{Sp}(C)$ is presentable and

$$\Omega^\infty : \mathbf{Sp}(C) \rightarrow C$$

admits a left adjoint $\Sigma_+^\infty : C \rightarrow \mathbf{Sp}(C)$. If C is pointed, we write this functor as Σ^∞ .

The important point is not merely the existence of this adjunction, but its universal property.

Theorem 4.5.8 (Free Stable Presentable Category). Let C and D be presentable and suppose that D is stable. Precomposition with Σ_+^∞ induces an equivalence

$$\mathbf{Fun}^L(\mathbf{Sp}(C), D) \xrightarrow{\sim} \mathbf{Fun}^L(C, D)$$

where $\mathbf{Fun}^L$ denotes the ∞ -category of colimit-preserving functors.

Taking $C = \mathbf{Ani}$ gives the fundamental characterization

$$\mathbf{Fun}^L(\mathbf{Sp}, D) \xrightarrow{\sim} D, \quad F \mapsto F(\mathbb{S})$$

Hence $\mathbf{Sp}$ is the stable presentable ∞ -category freely generated under colimits by one object, the sphere spectrum. This is the precise categorical sense in which $\mathbb{S}$ behaves like the free rank-one object $\mathbb{Z}$.

Remark. If $G : D \rightarrow \mathbf{Sp}(C)$ is exact, then G admits a left adjoint exactly when $\Omega^\infty G : D \rightarrow C$ does. Stabilization therefore does not create a new obstruction to adjointness.

4.5.4 Stable Localizations and Accessible t -Structures

Proposition 4.5.6 (Stable Localizations). Let C be stable and let $C' \subseteq C$ be a reflective full subcategory with localization $L : C \rightarrow C'$. Then C' is stable if and only if L is left exact.

Since a localization is a left adjoint, it already preserves colimits. Thus left exactness is precisely what makes it exact and allows stability to pass to the essential image.

Theorem 4.5.9 (Presentations by Spectral Presheaves). An ∞ -category C is presentable and stable if and only if there exist a small ∞ -category E and an accessible left-exact localization

$$\mathbf{Fun}(E^{\mathrm{op}}, \mathbf{Sp}) \rightarrow C$$

This is the stable analogue of presenting an ordinary Grothendieck category by a localization of a presheaf category: spaces are replaced by spectra, and left exactness preserves the stable structure.

Proposition 4.5.7 (Generating Accessible t -Structures). Let C be presentable and stable.

1. If $U \subseteq C$ is presentable and closed under small colimits and extensions, then there is a t -structure with $C_{\geq 0} = U$.
2. Every small family $\{x_\alpha\}$ generates an accessible t -structure: its connective part is the smallest full subcategory containing all x_α and closed under extensions and small colimits.

Remark (Accessibility Test). For a t -structure on a presentable stable ∞ -category, the following conditions are equivalent: $C_{\geq 0}$ is presentable, $C_{\leq 0}$ is presentable, and the truncation functors $\tau_{\geq 0}$ and $\tau_{\leq 0}$ are accessible. Any of these conditions is what we mean by an *accessible t -structure*.

4.6 Mapping Spectra

Mapping spaces in a stable ∞ -category admit compatible deloopings. A single spectrum therefore packages maps in every shifted degree.

4.6.1 Construction

Let C be a locally small stable ∞ -category and let $x, y \in C$. For $n \geq 0$, set

$$m_n := \mathrm{Map}_C(x, \Sigma^n y)$$

Since $\Sigma^n y \simeq \Omega \Sigma^{n+1} y$ and mapping out of x preserves limits, there are natural equivalences

$$m_n \simeq \Omega m_{n+1}$$

These equivalences, with their canonical coherences, define an Ω -spectrum.

Definition 4.6.10 (Mapping Spectrum). The *mapping spectrum* from x to y is the spectrum

$$\mathrm{Map}_C^{\mathbf{Sp}}(x, y) \in \mathbf{Sp}$$

whose n th space is $\mathrm{Map}_C(x, \Sigma^n y)$. Its infinite loop space is the original mapping space

$$\Omega^\infty \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \simeq \mathrm{Map}_C(x, y)$$

4.6.2 Basic Properties

Proposition 4.6.8. For $x, y \in C$ and $a, b, n \in \mathbb{Z}$, the mapping spectrum satisfies:

1. *Homotopy groups recover shifted maps.*

$$\pi_n \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \simeq \mathrm{Hom}_{hC}(\Sigma^n x, y) \simeq \mathrm{Hom}_{hC}(x, \Sigma^{-n} y)$$

2. *Shifts become suspensions of spectra.*

$$\mathrm{Map}_C^{\mathbf{Sp}}(\Sigma^a x, \Sigma^b y) \simeq \Sigma^{b-a} \mathrm{Map}_C^{\mathbf{Sp}}(x, y)$$

3. *Exactness.* The functors

$$\mathrm{Map}_C^{\mathbf{Sp}}(x, -) : C \rightarrow \mathbf{Sp}, \quad \mathrm{Map}_C^{\mathbf{Sp}}(-, y) : C^{\mathrm{op}} \rightarrow \mathbf{Sp}$$

are exact.

4. *Detection of equivalences.* A map $f : y \rightarrow z$ is an equivalence if and only if $\mathrm{Map}_C^{\mathbf{Sp}}(x, f)$ is an equivalence for every $x \in C$.

For example, a cofiber sequence $x \rightarrow y \rightarrow z$ induces fiber sequences

$$\mathrm{Map}_C^{\mathbf{Sp}}(w, x) \rightarrow \mathrm{Map}_C^{\mathbf{Sp}}(w, y) \rightarrow \mathrm{Map}_C^{\mathbf{Sp}}(w, z)$$

and

$$\mathrm{Map}_C^{\mathbf{Sp}}(z, w) \rightarrow \mathrm{Map}_C^{\mathbf{Sp}}(y, w) \rightarrow \mathrm{Map}_C^{\mathbf{Sp}}(x, w)$$

Taking homotopy groups recovers the familiar long exact sequences of Ext-groups. In the notation of Chapter 2,

$$\pi_n \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \simeq \mathrm{Ext}_C^{-n}(x, y)$$

Remark (Spectral Enrichment). Composition of maps refines canonically to maps of spectra

$$\mathrm{Map}_C^{\mathbf{Sp}}(y, z) \wedge \mathrm{Map}_C^{\mathbf{Sp}}(x, y) \rightarrow \mathrm{Map}_C^{\mathbf{Sp}}(x, z)$$

together with unit maps

$$\mathbb{S} \rightarrow \mathrm{Map}_C^{\mathbf{Sp}}(x, x)$$

These maps are coherently associative and unital. In other words, every stable ∞ -category is canonically enriched over spectra. Ordinary mapping spaces are obtained by applying Ω^∞ and therefore remember only the infinite-loop-space face of this richer object.

Remark (The Presentable Case). If C is presentable and stable, it is canonically tensored over $\mathbf{Sp}$. For $e \in \mathbf{Sp}$ and $x \in C$, there is an object $e \otimes x \in C$ characterized by the tensor–Hom adjunction

$$\mathrm{Map}_C^{\mathbf{Sp}}(e \otimes x, y) \simeq \mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}(e, \mathrm{Map}_C^{\mathbf{Sp}}(x, y))$$

The functor $e \mapsto e \otimes x$ preserves colimits and sends $\mathbb{S}$ to x . For $C = \mathbf{Sp}$, the mapping spectrum is the usual function spectrum $F(x, y)$, and

$$\mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}(\mathbb{S}, y) \simeq y$$

4.7 Cohomology and Homology Theories

Fix a spectrum $E \in \mathbf{Sp}$ and a pointed anima $x \in \mathbf{Ani}_*$. In formulas below, we use the abbreviations

$$E \wedge x := E \wedge \Sigma^\infty x, \quad \mathrm{Map}^{\mathbf{Sp}}(x, E) := \mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}(\Sigma^\infty x, E)$$

Thus the space x is first stabilized; smash products and mapping spectra are then formed inside $\mathbf{Sp}$.

4.7.1 Cohomology First

Definition 4.7.11 (Reduced Generalized Cohomology Theory). A *reduced generalized cohomology theory* on pointed anima is a family of contravariant functors

$$E^n : (h\mathbf{Ani}_*)^{\mathrm{op}} \rightarrow \mathbf{Ab} \quad (n \in \mathbb{Z})$$

satisfying the following axioms.

1. *Reducedness.*

$$E^n(*) = 0$$

2. *Suspension.* There are natural isomorphisms

$$E^n(\Sigma x) \simeq E^{n-1}(x)$$

3. *Exactness.* Every cofiber sequence

$$a \rightarrow x \rightarrow x/a \rightarrow \Sigma a$$

induces a natural long exact sequence

$$\dots \rightarrow E^n(x/a) \rightarrow E^n(x) \rightarrow E^n(a) \rightarrow E^{n+1}(x/a) \rightarrow \dots$$

4. *Wedge axiom.* For every family $\{x_\alpha\}$,

$$E^n\left(\bigvee_\alpha x_\alpha\right) \simeq \prod_\alpha E^n(x_\alpha)$$

The word *generalized* means that no dimension axiom is required.

Definition 4.7.12 (Representing Spectrum). Let $\{E^n\}_{n \in \mathbb{Z}}$ be a reduced generalized cohomology theory. A *representing spectrum* is a spectrum $E \in \mathbf{Sp}$ together with natural isomorphisms

$$E^n(x) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n E)$$

for every pointed anima x and every $n \in \mathbb{Z}$.

The more intrinsic object is not the separate family of abelian-group-valued functors E^n , but the single spectrum-valued functor

$$E^* : \mathbf{Sp}^{\mathrm{op}} \rightarrow \mathbf{Sp}, \quad y \mapsto \mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}(y, E)$$

The original theory on pointed anima is obtained by precomposing with the suspension-spectrum functor

$$\mathbf{Ani}_* \xrightarrow{\Sigma^\infty} \mathbf{Sp} \xrightarrow{\mathrm{Map}_{\mathbf{Sp}}^{\mathbf{Sp}}(-, E)} \mathbf{Sp}$$

and then taking homotopy groups. Explicitly,

$$E^n(x) := \pi_{-n} \mathrm{Map}^{\mathbf{Sp}}(x, E) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n E)$$

Thus all groups $E^n(x)$ are the homotopy groups of one mapping spectrum, not unrelated invariants in different degrees.

Proposition 4.7.9 (A Spectrum Represents Cohomology). Every spectrum E defines a reduced generalized cohomology theory by the formula above. The representing spectrum is unique up to equivalence.

Proof sketch. The suspension formula follows from the shift formula for mapping spectra. Mapping into E carries cofiber sequences to fiber sequences and wedges to products. Taking homotopy groups gives exactness and the wedge axiom. Uniqueness follows from the stable Yoneda lemma. $\square$

Remark (Brown Representability). Conversely, Brown representability says that a reduced generalized cohomology theory satisfying the usual size hypotheses is represented by a spectrum E . Thus a whole family $\{E^n\}_{n \in \mathbb{Z}}$ is encoded by one stable object rather than by unrelated functors in each degree.

4.7.2 Homology

Definition 4.7.13 (Reduced Generalized Homology Theory). A *reduced generalized homology theory* is a family of covariant functors

$$E_n : h\mathbf{Ani}_* \rightarrow \mathbf{Ab} \quad (n \in \mathbb{Z})$$

satisfying reducedness, natural suspension isomorphisms

$$E_n(\Sigma x) \simeq E_{n-1}(x)$$

exactness for cofiber sequences, and the wedge axiom

$$E_n\left(\bigvee_{\alpha} x_{\alpha}\right) \simeq \bigoplus_{\alpha} E_n(x_{\alpha})$$

Proposition 4.7.10 (Homology Represented by a Spectrum). Every spectrum E defines a reduced generalized homology theory by

$$E_n(x) := \pi_n(E \wedge x)$$

A cofiber sequence $a \rightarrow x \rightarrow x/a$ induces long exact sequences

$$\dots \rightarrow E_n(a) \rightarrow E_n(x) \rightarrow E_n(x/a) \rightarrow E_{n-1}(a) \rightarrow \dots$$

Proof sketch. Smashing with E preserves colimits, so it preserves cofiber sequences and wedges. Taking homotopy groups gives the long exact sequence and turns wedges of spectra into direct sums of abelian groups. $\square$

4.7.3 The Derived Algebra Dictionary

The analogy with derived algebra becomes literal after replacing the ground ring R by the sphere spectrum $\mathbb{S}$. A coefficient spectrum E behaves like a module, mapping spectra behave like derived Hom, and smash products behave like derived tensor products.

derived algebra	stable homotopy
$\mathrm{RHom}_R(m, n)$	$\mathrm{Map}^{\mathbf{Sp}}(x, E)$
$\mathrm{Ext}_R^q(m, n)$	$E^q(x)$
$m \otimes_R^{\mathbf{L}} n$	$x \wedge E$
$\mathrm{Tor}_q^R(m, n)$	$E_q(x)$

The two numerical theories are obtained by taking homotopy groups of the corresponding spectrum-valued constructions

$$E^q(x) = \pi_{-q} \mathrm{Map}^{\mathbf{Sp}}(x, E), \quad E_q(x) = \pi_q(x \wedge E)$$

Example (Ordinary and Stable Theories). For an abelian group a , the Eilenberg–Mac Lane spectrum Ha recovers ordinary reduced theories

$$(Ha)_n(x) \simeq \tilde{H}_n(x; a), \quad (Ha)^n(x) \simeq \tilde{H}^n(x; a)$$

Taking $E = \mathbb{S}$ instead gives stable homotopy and stable cohomotopy

$$\mathbb{S}_n(x) \simeq \pi_n(\Sigma^\infty x), \quad \mathbb{S}^n(x) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n \mathbb{S})$$

Remark. For an unpointed anima x , first adjoin a disjoint base point and use x_+ . Thus the unreduced theory is written

$$E_n(x) := \pi_n(E \wedge \Sigma_+^\infty x)$$

4.8 The Eilenberg–Mac Lane Spectrum

4.8.1 The Construction

Let $a \in \mathbf{Ab}$. For $n \geq 0$, let $a[n]$ denote the nonnegatively graded chain complex with a in degree n and zero elsewhere. Apply the Dold–Kan functor and forget the abelian-group structure to obtain a pointed anima

$$K(a, n) := \mathrm{DK}(a[n])$$

Every simplicial abelian group is a Kan complex, and Dold–Kan identifies homotopy groups with homology groups. Hence

$$\pi_i K(a, n) \simeq \begin{cases} a & \text{if } i = n \\ 0 & \text{if } i \neq n \end{cases}$$

Thus $K(a, n)$ is an Eilenberg–Mac Lane space. Shifting the chain complex corresponds to delooping, giving natural equivalences

$$K(a, n) \simeq \Omega K(a, n + 1)$$

These equivalences are coherent in n and therefore define an Ω -spectrum

$$K(a, 0), K(a, 1), K(a, 2), \dots$$

Definition 4.8.14 (Eilenberg–Mac Lane Spectrum). The spectrum determined by the sequence above is the *Eilenberg–Mac Lane spectrum* of a , denoted

$$Ha \in \mathbf{Sp}$$

Equivalently, Ha is the object corresponding to a under the equivalence

$$\mathbf{Sp}^\heartsuit \simeq \mathbf{Ab}$$

Proposition 4.8.11. The spectrum Ha is characterized up to equivalence by

$$\pi_i(Ha) \simeq \begin{cases} a & \text{if } i = 0 \\ 0 & \text{if } i \neq 0 \end{cases}$$

The construction is functorial and defines a fully faithful functor

$$H : \mathbf{Ab} \rightarrow \mathbf{Sp}$$

whose essential image is the heart $\mathbf{Sp}^\heartsuit$.

Proof sketch. The zeroth space of Ha is the discrete group $K(a, 0)$, and the equivalences $K(a, n) \simeq \Omega K(a, n + 1)$ identify the stable homotopy groups with the displayed Dold–Kan calculation. The description of the Postnikov heart then gives uniqueness, functoriality, and full faithfulness. $\square$

Remark (Why It Represents Ordinary Theory). For every pointed anima x , the general formulas represented by a spectrum specialize to

$$\tilde{H}^n(x; a) \simeq \mathrm{Hom}_{h\mathbf{Sp}}(\Sigma^\infty x, \Sigma^n Ha)$$

and

$$\tilde{H}_n(x; a) \simeq \pi_n(Ha \wedge \Sigma^\infty x)$$

Hence the Dold–Kan construction of the spaces $K(a, n)$, the spectrum Ha , and ordinary singular homology and cohomology are three levels of the same construction.

Remark. The spectrum $H\mathbb{Z}$ is a commutative ring spectrum, and every Ha is naturally an $H\mathbb{Z}$ -module. This is the stable-homotopical form of the ordinary $\mathbb{Z}$ -module structure on an abelian group. It also clarifies the earlier distinction: $H\mathbb{Z}$ encodes ordinary homological algebra, while the sphere spectrum $\mathbb{S}$ retains the higher stable stems.

5 ∞ -Operads

5.1 Motivations